Program overview

20-May-2025 12:55

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Robotics**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master Robotics 2024			
RO First Year (60 EC)			
RO Obligatory Courses (40 EC)			
RO47000	Vision and Reflection	0	
RO47001	Robot Dynamics & Control	5	
RO47002	Machine Learning for Robotics	5	
RO47003	Robot Software Practicals	5	
RO47004	Machine Perception	5	
RO47005	Planning & Decision Making	5	
RO47006	Human Robot Interaction	5	
RO47007	Multidisciplinary Project	5	
RO47008	Robot & Society	5	
RO Recommended elective courses on Robotics (at least 5 ECTS)			
RO47013	Control in Human-Robot Interaction	5	
RO47014	Knowledge Representation & Symbolic Reasoning	5	
RO47015	Applied Experimental Methods	5	
RO47017	Vehicle Dynamics and Control	5	
RO47018	Security and Privacy in Control	3	
RO47020	Advanced Machine Perception	5	
RO Recommended elective courses on Transferable skills (between 3 and 6 ECTS)			
AE4006E	Sustainability in Engineering	5	
AE4321-15	Air Traffic Management	4	
EPA131A	Behaviour in Transitions	5	
EPA143A	Macro-economics for Policy Analysis	5	
MT44070	Shipping Management	5	
TPM024A	Methods for Risk Analysis and Management	5	
TPM027A	Cyber Risk Management	5	
TPM033A	Sustainable Innovation and Transitions	3	
TPM0637SET	Economic Policy for Sustainable Energy	4	
TPM411A	Idea to Start-up IT & AI	5	
TPM416A	Turning Technology into Business	6	
TPM420A	Ready to Start-up	6	
WI4425	Financial Markets Theory	6	
WM0329TU	Ethics and Engineering	6	
WM0376TU	Ethics of Technological Risks	5	
WM0801TU	Introduction to Safety Science	3	
WM1302TU	Ethics of Transportation	5	
RO General elective courses			
AE4301	Automatic Flight Control System Design	3	
AE4316	Aerospace Human-Machine Systems	4	
AE4317	Autonomous Flight of Micro Air Vehicles	4	
BM41040	Neuromechanics & Motor Control	5	
CESE4025	Real-time Systems	5	
CESE4030	Embedded Systems Laboratory	5	
CS4205	Evolutionary Algorithms	5	
CS4235	Socio-Cognitive Engineering	5	
CS4295	Release Engineering for Machine Learning Applications	5	
EE4675	Object classification with radar	4	
ET4117	Electrical Machines and Drives	4	
ET4169	Radar I: From Basic Principles to Applications	5	
ET4257	Sensors and Actuators	4	
ME41006	Musculoskeletal Modeling and Simulation	4	
ME41056	Multibody Dynamics	5	
ME41065	System Identification and Parameter Estimation	7	
ME41085	Biomechatronics	4	
ME41096	Bio Inspired Design	5	
ME41125	Introduction to Engineering Research	3	
ME44206	Quantitative Methods for Logistics	5	
ME44312	Machine Learning for Transport and Multi-Machine Systems	3	
ME46055	Engineering Dynamics	4	
ME46060	Engineering Optimization: Concepts and Applications	3	
RO47019	Intelligent Control Systems	4	
		3	

SC42075	Modeling and Control of Hybrid Systems	
SC42101	Networked and Distributed Control Systems	4
SC42110	Dynamic Programming and Stochastic Control	5
SC42125	Model Predictive Control	4
WI4062TU	Transport, Routing and Scheduling	3
WI4156(TU)	Game theory	6
WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
WI4771TU	Object Oriented Scientific Programming with C++	3

RO Second Year

RO Electives 2nd year (Choose 1 out of 2, or 15 EC Free Electives)

IFM4040	Joint Interdisciplinary Project	15
RO57015	RO Internship / Research Assignment	15

RO Graduation Project Obligatory (45 ECTS)

RO57010-20	RO Literature Research	10
RO57010-20 toets 1	*Literature Research*	*10*
RO57010-20 toets 2	*Presentation*	*0*
RO57010-20 toets 3	*Colloquium Attendance*	*0*
RO57035	RO MSc Thesis	35

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Robotics**

Master Robotics 2024	
Director of Studies	J.F.P. Kooij
Contact for students	coordinator-cor-ME@tudelft.nl
Introduction 1	The Robotics programme focusses on four connected didactical goals: to give students an understanding of the development of intelligent robots and vehicles, which will advance mobility, productivity, and quality of life with a firm root in the theory and wide focus on applications; to train students to handle the entire process of innovative (sustainable) designing, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing; to coach students to perform research on robotic topics at an academic level; to teach students to act in a complex and multifunctional environment in different roles and in developing transferable skills. In addition to substantive knowledge of the different disciplines, students also gain experience with transferable skills, which are becoming increasingly important for engineers. Transferable skills are a core set of skills and abilities such as communication, teamwork, organisation and time management, problem-solving, leadership, and openness to change. Students get the opportunity to acquire these skills in multi- and interdisciplinary projects, in assignments, internships, and thesis work. Vision on the profession The Robotics Engineer will supervise the transition towards further robotisation of society. The Robotics Engineer masters a variety of mechanical engineering and computer science disciplines: mechatronics, artificial intelligence, embedded systems, control, human-machine systems, ethics, and security. The Robotics Engineer can solve complex and multi-functional robot problems, is a critical thinker beyond cultural boundaries, has a can-do mentality, and understands privacy, security, and ethical aspects of working with robots in human environments. Professional profile The professional profile is not yet clear because the Robotics field is developing quickly. Discussions with companies, research institutes, alumni, and scientific staff make clear that in a professional capacity, the Robotics Engineer is involved in: awareness of how applications work out in practice; translating societal issues into intelligent machine solutions in complex (multidimensional) situations that take into account ethics, safety, and sustainability; keeping up to date on technical developments in robotics and artificial intelligence; researching, developing, implementing, and testing artificial intelligence for mechanical engineering systems so that they can better learn and interact with their environment; developing mathematical models (perception models, behavioural models, situation analyses); physical modelling; programming and developping intelligent software for mechanical engineering systems; advising businesses, the government, and society on future choices and steps concerning the use and development of robotics; handling all information from sensors and integrating them in complex robot solutions.
Exit Qualifications	The graduated Master Robotics meets, to a sufficient level, the following qualifications: 1. Competent in the scientific discipline Robotics A graduate in Robotics is able to 1A. acquire and apply broad knowledge on Robotics on the multidisciplinary intersection of mechanical engineering, systems and control, and artificial intelligence, more particularly in dynamics, system identification, modelling, control, machine learning, machine perception, and human-robot interaction. 1B. model, design and control robotic systems. 1C. analyse, evaluate and validate robotic systems in complex environments. 1D. relate scientific knowledge to robotic systems, critically considering their interaction with societal aspects. 2. Competent in doing research A graduate in Robotics is able to 2A. study a topic by critically selecting relevant scientific literature. 2B. generate innovative contributions in developing intelligent machines. 2B. write a scientific report about own research. 2D. measure, model, and explain the interaction between humans and intelligent machines. 3. Competent in designing A graduate in Robotics is able to 3A. develop mathematical and physical systems using state-of-the-art knowledge. 3B. translate complex multidisciplinary research to working robotic designs. 3C. design algorithms and software for complex robots. 3D. design interfaces for human interaction so that a robotic systems functionality can be understood, taught and corrected by users. 3E. design robotic systems which can move safely and efficiently in human-inhabited environments. 4. A scientific approach A graduate in Robotics is able to 4A. contribute novel techniques on the intersection between mechanical engineering, systems and control, and artificial intelligence. 4B. analyse and design multidisciplinary solutions using system identification, modelling, and simulation. 4C. solve technological problems in a changing environment, considering ethics, safety, ambiguity, incompleteness and limitations. 4D. effectively lead, co-create and collaborate with a research team. 4E. design and perform experiments to compare, investigate, evaluate and test different robotic solutions across disciplines. 5. Basic intellectual skills A graduate in Robotics is able to 5A. develop a vision for applying robotics to address industrial and societal needs. 5B. consider the design of robotic systems from economic, social, cultural and ethical perspectives. 5C. critically reflect on own role in projects, in relation to that of others. 5D. remain professionally competent with an eye for the needs in the field. 6. Competent in operating and communicating A graduate in Robotics is able to 6A. work both independently as well as in a multidisciplinary team 6B. operate and communicate in a responsible, ethical and transparent manner, with an open-minded attitude. 6C. explain and defend research activities and outcomes to academia and industry, both specialists and non-specialists. 6D. understand and explain robotic systems in relation to other fields. 6E. ...advise on technical steps towards integral robotic solutions in complex environments.

	7. Considering the temporal and social context A graduate in Robotics is able to 7A. consider the limitations and possibilities of applying robotics to solving safe technological and societal problems. 7B. evaluate and assess the technological, ethical and societal impact of ones work. 7C. act with vision in an interconnected and rapidly changing world. 7D. act with integrity and responsibility regarding sustainability, safety and privacy, and economic and social wellbeing.
Program Structure 1	First year The first year of the programme consists of eight obligatory courses for all Robotics students, with a total of 40 EC. The goal of these courses is to give all students a solid common background in Robotics. The courses are (1) Dynamics and Control, (2) Machine Learning, (3) Robotics Practicals, (4) Machine Perception, (5) Planning and Decision Making, (6) Human-Robot Interaction, (7) Robots and Society, and (8) Multidisciplinary Project. Second year The second year of the programme consists of three elements: (1) a project with focus on interdisciplinarity. Students can choose between an Internship and a Joint Interdisciplinary Project (15 EC), (2) Literature Research (10 EC), and (3) Thesis (35 EC). The goal of the second year is to let students work on complex research problems both individually and in groups, apply the tools and methods taught in the first year, and develop new theories or design methods to solve complex mechanical engineering problems.

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RO First Year (60 EC)

Year	**2024/2025**
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RO Obligatory Courses (40 EC)

RO47000	Vision and Reflection	0
Responsible Instructor	Prof.dr.ir. D.A. Abbink	
Instructor	Ir. C.C. Claij	
Responsible for assignments	Dr.ir. O. Siebinga	
Co-responsible for assignments	Ir. T. Hoedemakers	
Contact Hours / Week x/x/x/x	nvt	
Education Period	1 2	
Start Education	1 2	
Exam Period	1 2	
Course Language	English	
Required for	All MSc Robotics students	
Course Contents	The modern robotics engineer is expected to have not only technical knowledge but also transferable skills, such as being able to work in teams, (self-)leadership, decision-making and communication skills, the ability to plan and prioritise goals and tasks, and the ability to reflect on experiences. The goal of RO47000 is to further engage you in your personal and professional development. You will learn what kind of robotics engineer you would like to be, set goals and make a development plan, commit to your plan, and adjust your goals and plan based on your experiences. Talking about how you do things and what motivates and demotivates you requires practice and interpersonal interaction. Therefore, you will be encouraged to engage in conversations and practice giving and receiving constructive feedback. At the end of this course in Q2, you will have laid the groundwork for being a reflective engineer now and in the future.	
Study Goals	By the end of this course, you will be able to: - Set and plan a personal development (technical and interpersonal skills) based on personal ambitions and vision - Provide and receive constructive feedback - Reflect on learning experiences and adapt your goals and personal development plan accordingly - Have open and honest conversations about personal development	
Education Method	The above skills are not easily learned from a book or instruction video but have to be learned through action (interaction, engagement, and experimentation). You will be part of a mentor group with 1416 Robotics students and two mentors. The mentors are trained to facilitate your group, resulting in a "train-the-trainer" educational method. The mentor groups are formed and facilitated to provide a training ground for you to have open and honest conversations about personal development, and to reflect on experiences in the Robotics MSc program.	
Assessment	The obligatory deliverables are: 1. An elevator pitch presentation in which you introduce yourself and your ambitions: who are you and what kind of engineer would you like to be? (Q1, introduction week). 2. Your presence at a workshop that is connected to the RO47006 course (Human-Robot Interaction), during which you will work together with your mentor group on an assignment (Q2). 3. A vision presentation with your goals and personal development plan to your mentor group and a discussion in your mentor group (Q2). You will have to complete all three above-mentioned activities at a sufficient level to pass the course. The grade is insufficient if any of the three activities are not completed. There is no resit.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Cognitive Robotics	
Contact	David Abbink (d.a.abbink@tudelft.nl)	

RO47001	Robot Dynamics & Control	5
Responsible Instructor	L. Peternel	
Instructor	B. Shyrokau	
Instructor	J.M. Prendergast	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	

Course Contents

This course provides the basics for deriving kinematic/dynamical models of a robot (such as a factory robotic arm, mobile robot, etc.) and for designing corresponding robot controllers. Students will learn how to derive the equations of motion and basic control equations themselves, and they will also learn how to use existing frameworks to do the same.

THEORY SUBJECTS

Introduction & Fundamentals
-Course layout, instructors, grade scheme, etc.
-Importance of the course and the big picture
-Connection to real-world
-Basic concepts/terms in robotics
-Methods to describe robot pose
-Numerical integration

Recap of Control Theory
-State-space model
-Root locus
-Bode plot
-PID control
-Concept of LQR and MPC control
-Reference to online materials

Endpoint Control for Robotic Manipulators
-Big picture + motivation
-Difference between position and torque-controlled robots
-Endpoint position control (with a PID controller)
-Endpoint impedance control (for position tracking tasks)
-Endpoint force control (for force tracking tasks)

Robotic Arm Kinematics
-Deriving forward kinematics and Jacobian (for a 2-DoF planar arm)
-Transformations between endpoint and joint space using Jacobian
-Kinematic singularities and their solution
-Kinematic redundancy and task priority control (for a 4-DoF planar arm)
-Kinematics of parallel robots and differences compared to serial robots

Robotic Arm Dynamics
-Rigid-body dynamics model
-Deriving dynamics model with Lagrange method for 2-DoF planar arm
-Using the dynamics model in the control
-Dynamic task priority control

3D Robotic Arm
-Deriving 3D kinematics using the Denavit-Hartenberg convention
-3D trajectory generation

Planning & Motion Control
-Orientation control
-Stability analysis of the controller

Kinematics & Dynamics of Mobile Robot / Automated Vehicle
-Forward kinematic models
-Wheel kinematic constraints
-Robot kinematic constraints
-Non-holonomic system
-Derivation of linear bicycle model
-Steady-state analysis

Dynamics & Control of Mobile Robot / Automated Vehicle
-Transient analysis
-Frequency response
-Path-following control

PRACTICAL ASSIGNMENTS (individual)

Assignment 1:
The goal of this assignment is to learn how to control a 2-DoF robotic arm in Python to follow a prescribed trajectory using kinematic and dynamic equations.

Assignment 2:
The goal of this assignment is to learn how to control a multi-DoF robotic manipulator (3D kinematics, 3D trajectory, orientation control).

Assignment 3:
The goal of this assignment is to learn how to implement a linear bicycle model, and calculate and discuss primary handling characteristics, poles, stability analysis, and feedback control (PID).

Study Goals

After completing this course, the students are able to:

-Derive the equations of motion for different types of robots
-Create joint-level and endpoint-level controllers (including trajectory following)
-Implement a simulation model in Python or Matlab
-Analyse the designed control systems and reflect on the design choices

Education Method

-Lectures & interactive sessions (4 hours per week)
-Practical assignments (homework)

Note: While attending lectures & interactive sessions is not mandatory, it is highly recommended. Assignments are not strictly mandatory, however, if you do not submit them you will lose the corresponding percentage of the final grade. That might mean that even if you pass the exam, you might not have a high enough final grade to pass the course.

Literature and Study Materials

Study materials
-Lecture slides
-Papers, books, videos, etc. (links provided in Brightspace)

Robotics textbooks (either one)
-Lynch, K. M., & Park, F. C. (2017). Modern robotics. Cambridge University Press.
-Craig, J. J. (2017). Introduction to robotics: mechanics and control. Pearson Education.
-Siciliano, B., Sciavicco, L., Villani, L., & Oriolo, G. (2010). Robotics: modelling, planning and control. Springer Science & Business Media.

Control theory textbooks (either one)
-Ogata, K. (2010). Modern control engineering. Prentice hall.
-Dorf, R. C., & Bishop, R. H. (2001). Modern Control Systems: Solutions Manual (Vol. 12). Prentice Hall.

Useful toolboxes
-Corke, Peter. Robotics Toolbox for Python.
-Foote, Tully. "tf: The transform library." 2013 IEEE Conference on Technologies for Practical Robot Applications (TePRA). IEEE, 2013.

Prerequisites

-Linear algebra (including matrix and vector operations)
-Calculus
-Linear system theory at the BSc level (e.g., simple differential equations, Laplace transforms)
-Basic programming experience at the BSc level (Python/Matlab)

Assessment

Number of assessments:
-Four (3 assignments, 1 exam), all based on individual work.

-The exam will be a computer/digital exam in ANS on campus.

Grade calculation:
-If and only if the exam grade is 5.5 or higher, the grade will consist of 30% of the average score of the assignments and 70% of the exam grade. The final grade must be 6.0 or higher to pass the course.

Assignment submission lateness criteria:
<1 hour: -1 point
<1 day: -2 points
<2 days: -4 points
<3 days: -6 points
>3 days: no points given

Note: Assignments and exam grades are only valid for the academic year they were obtained in.

Enrolment / Application

Registration:
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To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.

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Department

ME Department Cognitive Robotics

Contact

For general course inquiries: Luka Peternel (L.Peternel@tudelft.nl)
For Assignment 1 and related content: Luka Peternel (L.Peternel@tudelft.nl)
For Assignment 2 and related content: Micah Prendergast (J.M.Prendergast@tudelft.nl)
For Assignment 3 and related content: Barys Shyrokau (B.Shyrokau@tudelft.nl)

RO47002	Machine Learning for Robotics	5
Responsible Instructor	Dr.ir. J. Kober	
Instructor	Dr. H. Caesar	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides a broad overview of machine learning techniques and their practical application in robotics. Intro to Machine Learning and python - Machine Learning in Robotics - Understanding the goal of machine learning, fundamental problems, high-level overview. Visualizing your data - Supervised vs Unsupervised vs Reinforcement learning - Model-driven vs Data-driven: White box, Gray box, Black box - Prior knowledge vs Unstructured. Feature extraction, linear regression + Deep Learning. Interpretability Hands-on Machine Learning - Example machine learning project in Robotics - Binary classification, decision boundaries, using logistic regression/SVM (one parameter) - How to perform a classification experiment. Dataset splitting, learning curves, metrics, comparing results Regression & Data collection - Regression and Data Collection in Robotics - Regression methods, least squares fitting - Overfitting, cross validation, regularization - Collecting (noisy) data, labelling data, outliers - High-dimensional data, data augmentation - Hyper-parameter optimization (grid search vs random search) Classification - Classification in Robotics - Parametric vs Non-parametric classifiers - Logistic regression - Decision tree, forest - Bayesian classification: Bayes' rule, naive Bayes, Gaussian Mixture Model - k-nearest neighbour - SVM, kernel-SVM, dual problem - Multi-class classification, metrics (confusion matrix), class imbalance Unsupervised Learning - Unsupervised Learning in Robotics - Clustering: K-means, Gaussian Mixture Model, DBSCAN - Dimensionality reduction: PCA, Local Linear Embedding (LLE) Neural Networks - Neural Networks in Robotics - Multi-Layer Perceptron, gradient descent - Neural Networks, and deep learning - Vanishing gradient problem, DropOut, Optimizers Advanced machine learning - Outlook: Recurrent neural networks, convolutional neural networks - Outlook: Data augmentation, AutoEncoder - Outlook: Reinforcement learning	
Study Goals	Knowledge, insight, judgement and skills: After completing this course, students will be able to - explain the role of ML in robotics, and describe possible use cases, and when not to use ML - explain the main concepts in ML (overfitting, training and validation, classification vs regression, cross validation, regularization, parametrics vs. non-parametrics, supervised/unsupervised/reinforcement learning) - explain the principles of well-established ML methods (e.g. Logistic regression, SVM, RFs, Neural Networks) - collect and preprocess data (normalization, feature selection, augmentation, splitting) for ML - analyze a ML problem, consider available data and computational resources, and select the appropriate ML methods to apply - perform experiments, evaluate the results, and draw sound conclusions - present the findings of a ML experiment clearly in a report using graphs and scientific English - write python code to visualize data and implement ML methods using popular python frameworks, such as pandas, numpy, matplotlib, jupyter notebook, sklearn, keras Transferable and interpersonal skills: After completing this course, students will be able to - collaborate and coordinate with a lab partner on coding and reporting tasks	
Education Method	Contact hours - Flipped classroom: pre-recorded lectures (2 hours per week) + classroom discussion (2 hours per week) - Q&A on practica (2 hours per week) Formative feedback - Quizzes -- Individual -- On material of previous lecture(s) -- Via BrightSpace at set time slots - Practica -- Pairwise collaboration -- Submission via BrightSpace -- Counts for knock-out criteria -- No late submission -- No resit Deadlines - For each practicum, the deadline is approx. one week after the practicum is published.	
Literature and Study Materials	Study materials - Slides (released shortly before lectures) - Book: "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent	

	Systems" 3rd edition, Aurélien Géron Suggested literature - "Pattern Recognition and Machine Learning", C. Bishop - "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Hastie, Tibshirani, Friedman
Prerequisites	TU Delft BSc Mechanical Engineering curriculum, or equivalent. Especially familiarity with - Programming in python - Linear Algebra - Probability theory
Assessment	Final assignment (30%) - Pairwise collaboration - Late submission: -1 grade point per day - No resit - Deadline: Will be released after the 6th practicum, giving the students approx. two weeks to read the assignment, collect the necessary data, perform their experiments, and write their report Exam (70%) - Individual - Written closed-book exam - Exam in Q1, resit in Q2 Knock-out criteria - At least 4 practicum grades need to be a pass (of current or previous academic year) - Minimum grade for final assignment is a 5 (of current or previous academic year) - Minimum grade for the exam is a 5 - Final grade minimum is a 6
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics

RO47003	Robot Software Practicals	5
Responsible Instructor	J.F.P. Kooij	
Instructor	G.A. van der Hoorn	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming experience (e.g. python or matlab) at the BSc level, familiarity with the use of functions, variables, and control structures (for-loops, if-statements, etc.).	
Course Contents	This course teaches students the basics for creating and collaborating on academic robotics software. It aims to give students sufficient knowledge and experience such that they can proceed learning more about robotics programming by themselves as needed. The course does this by providing an overview of key technologies, and letting students develop basic skills often used in real world robotic software development. The main objective is to obtain hands-on experience - beginners level - with Linux, Git & Gitlab, C++, ROS, and integrate with common external libraries such as OpenCV and PCL. This course does not teach new robotics theory (e.g. perception, planning, control, etc.) which is covered in other courses. The allotted 5 ECTS means that students are expected to invest approximately 11 hours/week outside of contact hours. Be aware that active participation in all the lab and peer-review assignments is required, hence it is not recommended to do this course if you cannot commit to a regular time investment over the whole education period.	
Course Contents Continuation	Linux: Although Windows is the most prevalent operating system on desktop PCs, Linux became very popular for embedded systems such as robots. Developing for embedded Linux systems is most easily done in Linux itself, and this course aims to familiarize students with the use of Linux on the desktop and terminal. It includes the following topics: OS architecture, File system, Shell, Scripting. Git/gitlab: All exercises will be submitted using Git, a widely used version control system to manage and collaborate on coding projects. During the course, students will practice basic tasks with git, such as solving code conflicts, and merging the work of lab partners. Using an issue tracker and giving constructive feedback to peers is also an important aspect that will be practiced. C++: C++ is one of the most relevant programming languages for robotics. Being an object oriented language brings great advantages compared to its predecessor C. This practical gives students practical knowledge on C++ programming. The course encompasses the following topics: Introduction to C++, common compilation tools, basic programming in C++, classes and objects. ROS: The Robot Operating System (ROS) is a flexible framework for writing robot software. It is a collection of tools, libraries, and conventions that aim to simplify the task of creating complex and robust robot behaviour across a wide variety of robotic platforms. During the course, students will learn to use existing ROS tools, create their own software components in C++, integrate them into an application, and test their solutions using a physics simulator. others: You will also get some practice using external libraries in your robotics project, such as OpenCV or PCL. OpenCV is a computer vision library designed for computational efficiency and with a strong focus on real-time applications. The Point Cloud Library (PCL) is an open-source project for point cloud processing. Note that this course is *not* about understanding the methods or theory on which these libraries are based, but rather on integrating the provided functionality in novel software components.	
Study Goals	--------------------------------------- Knowledge, insight, judgment and skills --------------------------------------- After completing this course, students will be able to 1. use the command line on a Linux system, perform basic file operations and job management tasks, and automate tasks with shell scripts; 2. use Git to commit and integrate incremental code changes together with a lab partner, and resolve any code integration conflicts; 3. use a Gitlab server to exchange code changes, and use its issue tracker to provide and receive feedback on each other's code; 4. solve programming tasks in C++, compile the code using Make, debug compile-time, link-time and run-time problems; 5. create new C++ ROS packages, use common ROS development tools, and utilize the ROS API in their C++ programs; 6. investigate and integrate external libraries into their own ROS components, such as OpenCV or PCL; 7. add code comments and basic documentation to their code, find and interpret documentation of external libraries; ------------------------------------- Transferable and interpersonal skills ------------------------------------- 8. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback; 9. collaborate and coordinate with lab partner on coding tasks.	
Education Method	Lectures, which will introduce new course content (2 hours per week). Interactive Sessions will provide class wide feedback on past lab assignments, tutorials, quiz exercises, Q&A (2 hours per week) Practicum session, during which there will be TAs around to help you understand the lab assignments (2 hours per week) Robotics Software Practicals focuses on improving programming and collaborative software development skills. The main method to develop these skills is by gaining hands-on experience through lab assignments, and receiving feedback on the submitted work. Lectures will introduce the various topics, provide context and tips, and will discuss common pitfalls and solutions. Still, the nature of the assignments means that students are expected to invest sufficient time for self-study and practice outside the fixed lab and lecture contact hours. Lab assignment There are 4 obligatory lab assignments, each covering two weeks, namely 1. Linux, Git (Formative) 2. C++ programming (Formative) 3. ROS (Formative) 4. Final Assignment (Summative, demonstrates capacity to use Linux, Git, C++, and ROS). A manual will be provided for each lab assignment before the start of the first practicum session for the assignment. The manual will contain detailed instructions, guidelines, optional exercises, and an assignment that must be submitted to an online repository for your lab group.	
Literature and Study Materials	1. A Gentle Introduction to ROS, Jason M. OKane. http://www.cse.sc.edu/~jokane/agitr/ 2. Provided lab manuals, and lecture slides that will be made available on Brightspace	

Practical Guide	For the practicum assignments, bring a reasonably recent laptop with a x86-64 CPU (i.e. a normal" Intel or AMD CPU; but NOT the new MacBook Air with M2 chip, NOT a Chromebook, etc. These use different CPU architecture which means that the software we provide will not run! A MacBook with an Intel x86 CPU is fine). A laptop with an Nvidia GPU would make the robot simulations run even smoother. Please note the following lab rules: The first three lab assignments are made in groups of two, as you will learn to collaborate on code with your lab partner using gitlab. You can find a lab partner using the special Brightspace forum, but of course also by talking to others after the first lecture, or by looking for a Brightspace lab group with a single enrolled person. To enrol in the practicum, and therefore participate in the course, you and your partner *must* enrol on Brightspace in a lab group before the lab enrol deadline (this deadline will be announced in first lecture). If you plan to drop the course, inform the lab coordinator in time before the start of the next lab assignment, but while you are still enrolled, you should finish the ongoing lab assignment with your lab partner. The scheduled practicum sessions are intended for you to obtain help on the assignments from the TAs. Outside the practicum sessions, you can post questions on the TA forum on Brightspace. To submit your work, you and your partner must push your contributions to a provided gitlab repository. Note that using git and the gitlab webserver for code collaboration is part of the learning objectives of this course. You are responsible for testing the code that you hand-in via gitlab, so make sure that you have tested your work thoroughly, and that the correct version has been uploaded to your gitlab repository. For the first three lab assignments, there will be a peer-review task after the lab deadline has passed. You will learn during the course how to provide feedback, and how to use the gitlab issue tracker to manage issues and comments. The lectures will afterwards discuss positively peer-reviewed lab solutions with all students for additional feedback by a teacher. Active participation in each lab assignment is required to participate in the next assignment. Active participation entails that you submit your solution before the given deadline, and that you complete the required peer-review tasks. You may only use the gitlab code repository that the lab coordinator will create for your group. You are *NOT* allowed to put your coding homework on github.com, gitlab.com, bitbucket.com, google drive, dropbox, etc. (neither with public nor private settings!) TU Delft ethical guidelines apply to reports, code, etc. If your name is on it, you claim it is your work, do *not* copy parts of other student groups code. You are *not* allowed to share your code, solutions with others apart from your lab partner and TAs. Generally, do *not* show your groups work to other students. Plagiarism will be reported to exam committee.
Assessment	# Assessment overview Formative: First three lab assignments, peer-review tasks, discussion on common mistakes and solutions of past hand-in assignments during the lecture by teacher, multiple-choice practice questions. Summative: The final lab assignment, and digital exam. # Grade calculation You final course grade (minimum is a 6 to pass) determined by Final lab assignment (grade, weight 25%, minimum a 5.5 to pass), Digital exam (grade, weight 75%, minimum a 5.5 to pass) Any penalties (see Deadlines below) A resit will be available for the exam, but not the final lab assignment. # Lab assignments You must enroll for the obligatory lab assignments before the announced enrollment deadline (given during first lecture), so that the lab coordinator can fix the groups and setup everybodys code repositories in time. If you do not enroll, you cannot make the lab assignments, or make the exam. In the first three assignments you must collaborate with a lab partner. Your group will receive and give formative feedback through student peer-review (what goes well, what to improve). The fourth and final lab assignment is made individually, and you receive a grade (weights 25% for final grade), based on a rubric for the quality of the solution, code, documentation, and use of Git. This assignments requires you to apply all knowledge from the previous lab assignments to solve a well-defined robotics programming task. Your (groups) lab work is evaluated on the following three Lab Criteria (LC): - LC1: solution quality (does the solution solve the given task?) - LC2: code quality (are there any bugs or problems with the code?), and - LC3: documentation quality (is there a clear readme, does the code contain sensible comments?). # Deadlines, submission requirements, and penalties While most of the lab assignments are formative, your active participation is required and there are strict submission and peer-review requirements: - LC4: Each partner must fulfil each labs minimum submission requirements regarding use of Git, and Gitlab, file layout, and build instructions, before the lab deadline. Each assignments unique submission requirements will be listed in the corresponding lab manual. - LC5: You must participate in the peer-review tasks for the first three lab assignments, and complete these before the given peer-review deadlines. Failing to comply with these criteria will result in penalties on your final *course* grade: - After lab submission deadline: -1 grade point per day for not completing the assignments minimum submission requirements, with a maximum of -2 per assignment. Not enrolling in time before the lab automatically results in -2. - After peer review deadline: immediate -0.5 grade point for not fully completing the peer-review task (only applies to those enrolled for the lab). NB: - These criteria apply to both lab partners individually. If you comply with the criteria but your partner does not, only your partner will receive the penalty. - Penalties apply to your final course grade, so if you do not submit for one lab (-2 penalty) you could only complete the course with an 8 at best. - There are NO redos or extra assignment opportunities to remove a given penalty. You will have to redo the lab next year and comply with the deadlines then for a clean slate. # Exam There will be an individually-made digital exam, for which you will receive a grade (weights 75% for final grade, a minimum of 5.5 to pass). Questions are based on content discussed in the lectures, and on practical situations that you have encountered during the lab assignments. The exam will also require you to reflect on and refer to your own coding solutions for the final lab assignment, therefore to participate in the exam you *must* have participated in the final lab assignment. Practice questions will be provided before the exam so you can assess your own understanding first.
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on

	https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics

RO47004	Machine Perception	5

Responsible Instructor	Prof.dr. D. Gavrila
Practical Coordinator	R.M. Ensing
Contact Hours / Week x/x/x/x	0.6.0.0
Education Period	2
Start Education	2
Exam Period	2 3
Course Language	English
Course Contents	This course provides an overview of machine perception techniques in robotics, with a focus on intelligent vehicles. Interested students outside ME (e.g. EEMCS, Civil Engineering and Aerospace Engineering faculties) with the proper background (see Prerequisites below) are encouraged to attend. Introduction Machine perception in robotics Course organization Sensor Overview (camera, radar, LiDAR, tactile) 3D Machine Vision Perspective camera model Extrinsic and intrinsic camera transformations Stereo vision Radar Sensing and Processing Principles of FMCW Radar Distance, velocity and angle measurements Use in object detection Lidar Sensing and Processing Point cloud representations Use in object detection Tactile Sensing and Processing Piezoresistive, capacitive, piezoelectric and optical sensing Contact and slippage Shape extraction and elasticity Visual Object Detection and Classification Detection vs. classification Object proposals Handcrafted features (e.g. HOG) & classification (e.g. linear SVM) End-to-end learning: neural networks, deep learning Performance metrics: confusion matrices, precision vs. recall, ROC curves State Estimation Bayesian Filtering Kalman Filtering Particle Filtering Object Tracking Data Association Track Management Self-Localization & Sensor Fusion Absolute vs. relative localization Ego-motion compensation (e.g. odometry, ICP algorithm) Extrinsic sensor calibration Environment representations (grids, voxels) The course has a substantial practicum component (about 65% of course work load), where learned concepts are put into practice by means of programming assignments (Python, using Jupyter notebooks).
Study Goals	After completing this course, students will be able to: explain the role of Machine Perception (MP) in Robotics, and describe possible applications explain the measurement principles of the relevant sensors explain the principles of well-established methods for low- to high-level sensor processing analyze a MP problem, consider available sensor and computational resources, and select the appropriate MP methods to apply write Python code in relevant frameworks to visualize data and implement MP methods perform MP experiments, evaluate the results, and draw sound conclusions
Education Method	Lectures (2-4 hours per week) Practicum (2 hours per week) Formative feedback: Practica - Pairwise collaboration (alternating random groups) - Pass/Fail (counts for knock-out criteria) - No late submissions or resit (except due to motivated medical conditions) Students can in principle run the software locally on their own computers (laptop/PC). However, we provide no support for setting up the necessary software environment (e.g. Anaconda). If the software environment cannot be installed properly (e.g. a few processors like M1 and ARM might not work), the student will need to revert to using the computers in the designated lab rooms.
Course Relations	This course, RO47004 Machine Perception *cannot* be selected in combination with ME41106 Intelligent Vehicles, due to the content overlap between the courses. This course, RO47004 Machine Perception provides the basics for the follow-up course "RO47020 Advanced Machine Perception".
Literature and Study Materials	Slides Hand-outs Jupyter notebooks
Prerequisites	Basic linear algebra and probability theory Intermediate Python programming skills, e.g. Numpy, object oriented programming (classes, inheritance), modules, namespaces and scope. Recommended: Pattern Recognition / Machine Learning (RO47002, CSE2510, or equivalent) It is possible to start the course with beginner Python skills if your aim is to take advantage of this course to also improve your programming expertise (a laudable aim, given its importance for both industry and academia). Be aware, however, that this

	catching up will likely result in an increased study load (e.g. 1-2 EC).
Assessment	Final practicum assignment (50%) - Individual assignment - Late submission: max. one day late for -1.0 grade point - Students having an initial grade of < 5.8 will be offered an opportunity for a remake to obtain a (maximum) grade of 5.8 for this component in the Q3 quarter. Final written exam (50%) - Written exam on campus. Knock-out criteria - At most one of the Pass/Fail practicum assignments is Fail - The minimum final practicum assignment grade is 5 - The minimum final written exam grade is 5 - The minimum final course grade is 6 The final practicum assignment will be evaluated on the computers in the designated lab rooms. If a student chooses to work using own computing equipment, the student needs to test the software on the designated computers before handing-in. Both final practicum assignment and final written exam results remain valid for one year. Each result can separately be transferred to the next academic year, upon request.
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Special Information	For more information about this course, see the slides of the Introduction lecture from past academic year at http://intelligent-vehicles.org/teaching/courses/ . TU Delft students might also be able to view the lectures from past academic year on Collegerama.
Department	ME Department Cognitive Robotics

RO47005	Planning & Decision Making	5
Responsible Instructor	Dr. J. Alonso-Mora	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an overview of motion planning and decision-making techniques and their practical application in robotics. Planning and Decision-making are critical components of autonomy in robotic systems. These components are responsible for making decisions that range from path planning and motion planning to coverage and task planning to taking actions that help robots understand the world around them better. This course studies underlying algorithmic techniques used for planning and decision-making in robotics and examines case studies in ground and aerial robots, mobile manipulation platforms and multi-robot systems. The students will learn the algorithms and implement them in a series of programming-based projects. In particular, we will first cover the fundamentals of planning (workspace, configuration space, representation of obstacles and robot models). We will then cover a broad set of methods, which include reactive methods and feedback control; discrete, combinatorial and probabilistic planning; planning under differential constraints; planning under uncertainty; learning in planning; and planning for multi-robot systems. Finally, we will briefly touch upon task assignment and vehicle routing. Course structure: Introduction Fundamentals: 3D rotations, configuration space, algorithm properties Differential constraints: recap of modeling & control of manipulators, ground robots, aerial robots Graph-search fundamentals (depth/breadth first, Dijkstra, A*, heuristics) Combinatorial algorithms (cell decomposition, shortest-path roadmaps, motion primitives) Sampling-based algorithms (PRM, RRT, RRT*) Reactive algorithms: collision avoidance, potential fields, velocity obstacles Kinodynamic planning Trajectory optimization (fundamentals, global, local, Model Predictive Control) Planning under uncertainty (Markov Decision Processes, learning) Multi-robot motion planning (joint configuration, optimization-based, coverage) Task assignment (Hungarian, auction, linear program) Intro to vehicle routing	
Study Goals	After completing this course, students will be able to: explain the role of Motion Planning and Decision-Making (PDM) in Robotics, and describe possible applications identify different classes of robotic systems, the associated mathematical models for their kinematics, dynamics and motion and the relevant spaces for PDM, such as the configuration and state spaces describe and compare algorithms for motion planning and decision-making of mobile robots and multi-robot systems, including: reactive algorithms, optimization-based algorithms, combinatorial algorithms, sampling-based algorithms and learning-based methods analyze a PDM problem, consider constraints, objectives and select the appropriate PDM methods to apply design and implement motion planners of moderate complexity to solve motion-planning tasks and navigate mobile robots perform PDM experiments, evaluate the results, and draw sound conclusions present the findings of a PDM experiment clearly in a report using graphs and scientific English understand PDM experiments descriptions found in scientific literature, and reproduce results	
Education Method	Lecture hours/week: 4 Q&A sessions/week: 2 Practical assignments and project (home) hours/week: 6 Self-study hours/week: 4	
Books	Planning Algorithms, S. LaValle http://planning.cs.uiuc.edu/ And additional papers/references provided by the lecturer.	
Prerequisites	Linear Algebra Programming in Python Rigid body dynamics (recommended) Dynamics & Control (recommended) Probability Theory (recommended)	
Assessment	Practicum assignments (10%) Project (40%) Collaboration in groups of 3 students - No resit (repeat the following year) Written Exam (50%) individual Knock-out criteria - Minimum grade for the project is a 5.0 - Minimum grade for the exam is a 5.0	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Cognitive Robotics	

RO47006	Human Robot Interaction	5
Responsible Instructor	Prof.dr.ir. J.C.F. de Winter	
Instructor	Prof.dr.ir. D.A. Abbink	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course on Human-Robot Interaction aims to equip students with the knowledge necessary to design robots that can function safely, comfortably, and effectively in environments inhabited by humans. The course offers both theoretical and empirical insights into the physical and cognitive aspects of human-robot interaction. Students will learn to model the fundamental principles of this interaction, critically examine existing theories and models, and assess various design options by reflecting on and experiencing experimental research.	
Course Contents Continuation	The course will cover key principles of human information-processing, human error, the interaction between humans and automation, human-machine interfaces, and operator state assessment (which includes factors like visual attention, workload, and situation awareness). The lectures will analyse the performance of human operators in roles ranging from manual control (direct control) and supervisory control of automated systems, to shared control of human-robot systems. Additionally, students will learn essential elements of experimental design for human-subject studies. The course finds its main applications in areas such as intelligent vehicles and industrial robotics. Additionally, guest lecturers from the departmernt will address specialized topics such as cognitive modeling for automated vehicles, haptic interfaces, and robotic neurorehablitation.	
Study Goals	Upon completing this course, students will be able to: - Explain the mechanisms of human information processing, which includes sensation, perception, decision-making, and action execution. - Explain the key theories and models related to human error, learning, and individual differences. - Describe and model direct human interactions with robots and other machines. - Articulate how humans supervise and interact with robots at higher levels of automation. - Critically examine concepts such as trust, overreliance, and behavioural adaptation in human-robot interaction. - Describe and reflect on the design of interfaces that facilitate effective human-robot cooperation. This includes classical interface design principles, but also more modern haptic interfaces as well as linguistic interfaces such as ChatGPT. - Adapt robotic functionality to human needs, either by teaching the robot or through operator state assessment. - Process experimental data and bio-signals to infer human visual attention, workload, and situation awareness. - Design and critically evaluate experimental studies involving human subjects for the assessment of robotic systems. - Provide constructive feedback to peers.	
Education Method	The course is taught through lectures, totalling four hours per week. Some of the lecture hours will be practical or interactive in nature. In addition to the regular lectures, there will also be guest lectures. Participation in sessions related to practical exercises is obligatory. Students will also participate in an experiment.	
Literature and Study Materials	The course materials will be made available on Brightspace. This includes: (1) the slides, (2) notes accompanying the lectures, and (3) scientific articles and book chapters.	
Assessment	Exam (45% of the final grade): This will be an on-campus digital examination. Additionally, the course incorporates two homework assignments: Assignment 1 is a group task related to manual control, while Assignment 2 is an individual task related to supervisory control of robotic systems and human-state assessment. Furthermore, students will participate in a human-subject experiment.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses To participate in midterms and endterms, timely registration for exams is obligatory. More information regarding registration for exams, the procedure, and deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations	
Department	ME Department Cognitive Robotics	

RO47007	Multidisciplinary Project	5
Responsible Instructor	Dr. C. Pek	
Instructor	Dr.ir. G.N. Saunders	
Instructor	Prof.dr.ir. M. Wisse	
Instructor	C. Hernandez Corbato	
Instructor	Ir. C.C. Claij	
Contact Hours / Week x/x/x/x	0.0.0.2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In this capstone project of the first year of the MSc Robotics, you will be challenged to use all the knowledge and skills you have gathered so far. Within a multidisciplinary team of up to 5 people, you will develop and test software for use in a real robot whilst working together for a real client in an industrial setting. We expect all your robotics skills (including your programming skills) to be up to date! The estimated workload of this course is 18 hours per week over 8 weeks. In this project, you will not just produce a software solution for a robot. You will also implement and test your software in a real robot and report and present your results as well as document your code, mirroring industry practice. Just like you would in industry, you will likely spend only 50% of your time coding and testing and the other 50% on meetings, systems design, report writing, (preparing) presentations, and documenting your code. These are no longer afterthoughts. In this project, you work closely with an industry client, realistically a possible future employer. That means there are expectations regarding your commitment and presence during the project. We expect you to behave cordially and respectfully towards the client and be on time for scheduled sessions. You will be given the option to express a preference for a client when you enroll if multiple clients take part in the course. Each of you will sign up with a specialist role in mind within the project, in which you are responsible for a specific part of the software design and implementation. It is no longer possible for one person to do all the programming. The specializations for which you can express a preference when you enroll are: - human-robot interaction, - navigation, planning, control - perception, and relate to the different disciplines within robotics. Within your team, you are expected to make a translation of societal issues from different perspectives (human, sustainability, safety, ethics, economic, etc. into intelligent robot solutions. Practically, each student team develops a complete, integrated software package for a complex robot system. You develop and test in a simulated environment, and then implement it in a real robot during the course before presenting your results in the final presentation to your client at the end. This course consists of 5 iterative phases with continuous reporting varying in lengths: - Phase 0: Introduction, meet your team, and get to know your robot. - Phase 1: Initiation and Planning - Getting organized, defining and planning your project. - Phase 2: Design & Execution - Software architecture design, initial (simulation) implementations and demonstration of concept to the client. - Phase 3: Implementation - Further design and implementation of your solution on your robot. - Phase 4: Testing & Completion - Finalizing your project and final reporting and code documentation, with a presentation of your implementation for the client in week 4.8. A detailed schedule will be available on Brightspace after the introduction lecture, likely in week 3.8.	
Study Goals	The learning objectives for this course are divided over two domains: 1) Knowledge, insight, judgment and skills By the end of the course, you should be able to: LO 1: Define a robotics use case with its functional and non-functional requirements. LO 2: Design robotic solutions by integrating knowledge of system engineering, current opportunities for robots, trends in robotics, and societal aspects. LO 3: Produce a robotic solution that meets the design specifications and considers sustainable, safety, ethical, economical and other relevant societal aspects. LO 4: Use functional architecture for planning and communicating robot software. LO 5: Communicate the multidisciplinary robot solution orally, in writing, and in code documentation according to robot software development guidelines. 2) Transferable and interpersonal skills. By the end of the course, you should be able to: LO 1: Formulate learning goals for your own personal development. LO 2: Reflect on your competencies (e.g., Teamwork, Leadership, Entrepreneurial thinking, Strategic multidisciplinary problem-solving), their development, and your personal learning goals. LO 3: Apply structured feedback to improve your performance or the performance of others. LO 4: Apply structured multidisciplinary software project management, with the use of different team roles and responsibilities.	
Education Method	Project work in teams supported by (video) lectures and workshops.	
Computer Use	During the project, we will be using different types of software and collaboration platforms. To save time and frustration, please ensure you have a working laptop that has the following software installed and have gained access to the following repository before the project starts: ROS: We expect you to have a working ROS environment on your laptop. Please use your knowledge and experience from RO47003 Robot Software Practicals and make sure that you align with the Ubuntu and ROS versions used in your clients robot. We will be using native installations of Ubuntu and ROS, and do not provide support for any other installation. Instructions on this will be published on Brightspace. Gitlab: All code and documentation for this project will be stored on GitLab. To be added to your groups repository on the GitLab environment for this course, go to: https://gitlab.tudelft.nl and log in using your NetID. Please do this as soon as you have been admitted to the course and have been assigned a group. Otherwise, you will not be able to access the required code for your project.	
Literature and Study Materials	During the project we expect you to use and refer to the following books. Please avail yourself prior to the course starting of either a hardcopy or an electronic copy of the book. [1] A Gentle Introduction to ROS Jason OKane, E-Book: Accessible via the authors website: https://jokane.net/agitr/ [2] Practical Model-Based Systems Engineering Jose Luis Fernandez & Carlos Hernandez Corbato, E-book: Accessible with the TU Delft library web extension In addition, you may also want to refer to these books. These are, however, not mandatory for the course: [3] Beginning Software Engineering Rod Stephens, Hardcopy: Available via Leeghwater, E-book: available through the TU Delft library extension. [4] Information Technology Project Management K. Schwalbe, Hardcopy: any online bookshop, E-Book: no free legal website available	
Prerequisites	Starting the project is only allowed when a student has successfully completed the course RO47003 "Robot Software Practicals" and has completed RO47000 Vision & Reflection. Without these preliminaries, the student will not be allowed to participate. Exemptions require written approval of the teachers of RO47003 and RO47000 that the student has obtained equivalent	

	qualifications. It is expected that you can code when you start this course.
Assessment	At the end of the course, your project needs to comply with a few requirements before we will grade it. These will be clearly listed in the course manual on Brightspace. Moreover, you must attend all mandatory sessions of the course. Excused absences are only permitted by receiving prior approval of a study advisor and the course staff. Absences due to medical reasons require a confirmation, e.g., by a study advisor. For this project, an individual grade is given. The grade for this project is a combination of the group grades for your final project including your final presentation, the Sprint assignments, and an individual grade based on your personal reflection. An individual bonus-malus of max. 1 grade point can be applied by staff in rare cases of excellence or underperformance. For more details on grading, see the course manual on Brightspace.
Enrolment / Application	Enrollment for the course will open at the beginning of Q3. Further details on how to enroll will be announced on Brightspace. All students must have registered before the announced registration deadline. It is the students responsibility to acquire the required registration information and deadline. After the registration deadline, late participation in the project is not possible anymore. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Special Information	Note: The mandatory introduction lecture and the ROS software & robot programming sessions will take place at the end of Q3, likely in week 3.8.
Remarks	This is a very intensive course which requires you to work 18 hours per week over a period of 8 weeks regardless of Public Holidays. The course requires intensive teamwork and good time management. On average there is 1 mandatory session per week you must be available for which can vary over the weeks depending on the availability of robots and industry clients. Past experience has shown that combining this course with too many electives and/or other intensive activities during weekdays is incompatible. Be prepared you will likely spend 40h per week on this project and any electives combined. It is also not recommended to start with your literature research during MDP.
Department	ME Department Cognitive Robotics

RO47008	Robot & Society	5
Responsible Instructor	O. Kudina	
Contact Hours / Week x/x/x/x	x/x/2/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Introduction Robots and Society takes a critical look at the design, development, production, and implementation of robots and AI in society. In the course, we will be examining many case studies as concrete applications of robotics and AI in society, e.g. healthcare robots, synthetic media in entertainment, decision-support systems in policing, human values when robots are concerned, as well as touch on the political and environmental aspects of robotics and AI. The course will provide students with the skills of critical reflection, ethical sensitivity and deliberation about the societal, design, environmental and governance issues at stake concerning these technologies. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for their future careers in the technical or humanities disciplines. The course serves as a first step in uncovering societal issues related to robotics and AI, understanding how and why they are points of concern, and applying conceptual tools to mitigate and/or overcome such issues. To participate in the course and be able to access the Course manual and all the materials, you simply need to enroll on Brightspace with the course code RO47008. Detailed content: Introduction to ethics of technology Technology as a sociotechnical system Practice-based approach to values Technological mediation of morality Value change Introduction to robot ethics Ethics of Human-Robot Interaction (e.g. social script, anthropomorphism, gender, inclusivity, etc.) Ethics of Robot Applications (e.g. healthcare robots, Digital Voice Assistants, military drones, etc.) Ethics of Robot Design and Development (e.g. Design for Values, Value hierarchy, Ethics as an accompaniment model, etc.) Introduction to ethics of AI Cultural and institutional aspects of AI Philosophical perspective on bias Power and politics of AI Ethics of AI in public sectors (e.g. predictive policing, facial recognition, Covid19 tracking apps, etc.)	
Study Goals	Learning goals: After completing the course, the students: are aware of the range of views on the relationship between robots, AI and society, from technology being neutral to technology being value laden to the technology as a sociotechnical system, can identify ethical issues and opportunities concerning specific robot and AI applications in society, have awareness for the specific ethical concepts and tools that can deployed for the responsible design of robotic and AI systems, such as Value Sensitive Design, technological mediation approach and the sociotechnical systems perspective; are capable of formulating a position on the ethical issues at stake in current applications of robotics and AI, develop evaluative judgments on the quality of their own work and of the performance of others (including peers) by implementing and providing feedback.	
Education Method	Contact hours: On-campus lectures (seven) and on-campus tutorials (four), weekly quizzes based on the readings and lectures, an exam, and a final paper. See the course schedule in MyTimetable. Weekly quizzes: Weekly quizzes will assess the students ability to understand the reading and lecture material from one week to the next and will prepare them for the exam. They are mandatory, have unlimited attempts and will act as a gateway to access the course materials for each upcoming week on Brightspace. Tutorials: Students are expected to attend four tutorials (A, B, C, D) (on campus). Each tutorial will have a different robotics or AI-related case study to analyze in the form of a debate, workshop, stakeholder group, etc. Prior to each tutorial, the students are expected to read/watch/listen several journalistic engagements and consider several questions, all to be discussed during the tutorial. Grade calculation: 60% of the grade will be based on the Exam. 40% of the grade will be based on the Final Paper. The minimum passing grade for each assignment is 5.5; however, to pass the course, your cumulative grade should be 6.0 or higher. Grades will be rounded to 0.5 (e.g. 7.2=>7.0 and 7.3=>7,5).	
Literature and Study Materials	 Study materials: Readings (available on Brightspace). Become available for each week after passing a quiz from the previous week. during the course. Lecture slides, available on Brightspace after the lecture. Quizzes, available before the lectures . Suggested literature: Nyholm, S. (2020). Humans and robots: Ethics, agency, and anthropomorphism. Rowman & Littlefield Publishers. Van de Poel, I. (2018). Design for value change. Ethics and Information Technology, 1-5. Wynsberghe, A. V. (2012). Designing Robots for Care: Care Centered Value-Sensitive Design. Science and Engineering Ethics, 19(2). Kudina, O., & Verbeek, P. P. (2019). Ethics from within: Google Glass, the Collingridge dilemma, and the mediated value of privacy. Science, Technology, & Human Values, 44(2), 291-314. Diakopoulos, N. (2020). Transparency. In Dubber, M., Pasquale, F., & Das, S. (Eds.) Oxford Handbook on AI Ethics (pp. 197-213). New York, NY: Oxford University Press.	
Assessment	1. Lectures and obligatory readings Each week, you need to read the mandatory readings for the Week to assist you in processing the lectures and preparing for the tutorials. All the materials are available on Brightspace, with lectures being on campus and recordings and pdfs becoming	

	available after each lecture. Students agree to use the reading material only for educational purposes and not share it beyond the course. You are encouraged to raise lecture- and reading-prompted questions on Brightspace and react to the posts of other students to enrich your learning with regards to the course. 2. Tutorials & Discussion Posts You must attend 4 on-campus tutorial sessions corresponding to tutorials A, B, C and D to complete the course. During each tutorial, you will complete a reflective exercise (e.g. a debate) and produce an analytic piece (e.g. a presentation) on a specific case of living well with robots in society. Each tutorial (A-D) is repeated over six sessions in two weeks to make sure you can have meaningful discussions in class and find the timeslot the suits your schedule best. Therefore, each tutorial session will accommodate typically 20-25 students. To get the spot in the session convenient to you (e.g. A4), consult the schedule and enroll into the groups of your choice already in Week 1 (see Brightspace Tutorial group enrollment). All tutorials will feature not just theoretical discussions, but material enactments of living with the robots, starting with the first tutorial, where you will be asked to build one by yourself. Each tutorial will have a designated subject and a series of preparatory materials (journalistic engagements, short videos and podcasts, etc.). A very important requirement the robot you will build in Tutorial A will need to accompany you = physically be there with you at every next tutorial session they were carefully designed by the teachers in an interconnected logic. So dont forget to bring your robots to all of the tutorials. To reiterate, attending four tutorials is mandatory and a prerequisite to passing the course. 3. Exam The exam will be held on campus in a computer room. It will be an open-book exam, where you will be able to access any course materials (provided that you cite and reference them, including access to the Internet and AI tools). You will not be allowed to consult or help your fellow students. You are expected to register for the exam in advance. More information on the exam will be in the Course manual. 4. Final Paper for the students on the 5ECTS trajectory: Individual reflective essay. You will be asked to reflect on the overarching question of this course, How to live well with robots?, using any course material (e.g. anything youve heard or read in the lectures, readings, tutorials) and crucially, reflecting on your experiences with the robot you will design during this course. In the essay, you will be asked to (1) identify any specific ethical issue raised by living with your robot (supported by your documented evidence) and inspired by it, to (2) anticipate any potential ethical issues in specific human-robot practices in society. You will be asked to (3) evaluate these problems, drawing on any course material, and to (4) make engineering recommendations on how to develop or apply robots a responsible way. Building on this, in summary, you will be asked to (5) answer the main courses question and give your recipe of what it takes to live well with robots. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics

RO Recommended elective courses on Robotics (at least 5 ECTS)

RO47013	Control in Human-Robot Interaction	5
Responsible Instructor	Dr. M. Wiertlewski	
Instructor	L. Peternel	
Contact Hours / Week x/x/x/x	0.0.6.0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	

Course Contents

Robots are increasingly pervasive in our lives and need to safely interact with humans in unstructured environments. This course covers the design principles and control strategies to ensure a safe and intuitive interaction between humans and these collaborative robots. A particular attention is put to the fundamentals needed to build and control robotic systems that physically interact with humans.

The course is split between interactive theory lectures and hands-on practical assignments using haptic interfaces. During the practical assignments, the theory is applied: first during a guided scenario, then with an open-ended mini-project. The first two assignments will teach you how to program real-time haptic rendering and to implement a teleoperation loop. The mini-projet will cover how to implement haptic guidance and validate the approach with human subjects.

After the practical assignments, additional lectures expand the perspective beyond the fundamental theory to industrial and consumer oriented applications. The course ends on an overview of the current state-of-the-art and remaining open challenges of the human-robot interaction field.

THEORY SUBJECTS

Fundamentals of physical interaction
- Motivation and relevance
- Existing examples

Impedance control
- Motivation behind modulating impedance of a robot
- Impedance and admittance control principles
- Hybrid force/impedance controller design
- Manipulability and its impact on interaction
- Implemented examples

Teleoperation
- Applications: when and why?
- Types: unilateral and bilateral
- Scaling position commands and force feedback

Tele-impedance
- Actively adapt robot impedance to achieve the desired interaction
- Impedance-command interfaces
- Implemented examples

Force feedback with virtual environments
- Providing human with physical virtual sensations
- Haptic rendering of virtual impedances
- Contact and Friction

Control of haptic interfaces
- Study of the stability with the Virtual Wall problem
- Effect of sampling on stability
- Transparency and Z-width to evaluate performance of haptic devices

Surface haptics (touch panel applications)
- Vibrotactile
- Ultrasonic
- Electroadhesion

Evaluation of human robot interaction
- Measuring performance with Fitts and Steering Law
- Learning effects with haptic guidance

Ergonomics in human-robot collaboration
- Basic human biomechanics
- Ergonomic metrics (manipulability, joint torque, muscle fatigue)
- Implemented examples

Haptic guidance
- Virtual fixtures
- Dynamic guidance
- Shared control

Mechatronics of HRI devices
- Sensors, actuators and linkages
- Designing the mechanical impedance
- Trade-off between performance and workspace
- Serial elastic actuators and soft robotics

Future outlook
- State-of-the-art research
- Emerging technologies
- Conclusion

PRACTICAL ASSIGNMENTS

Assignment 1: Haptic rendering (individual)
- Implement virtual environment
- Implement impedance/admittance control
- Observe and analyze stability issues

Assignment 2: Teleoperation (individual)
- Design a hybrid force/impedance controller
- Derive and visualize manipulability ellipsoid
- Implement teleoperation and tele-impedance
- Create a real-time UDP communication

Assignment 3: mini-project (individual or by group of 2)
Design and build a scenario to test the effect of haptic guidance on operators performance
Analyze the results and evaluate the performance of haptic guidance
Present the results to peers (and give feedback)

Study Goals

After this course, students will be able to:
- explain impedance-based robot control theory and its advantages
- apply the theory to different robots and tasks
- analyze the implementation issues and control performance
- create haptic feedback schemes to enhance human performance

Education Method

CONTACT HOURS
- Lectures (2 hours per week on average)
- Practical assignments (2 hours per week on average)

FEEDBACK
- Practical assignments (40%) (no resit) + individual reports
- Exam (60%) individual
- Homework individual self progress check

Knock-out criteria:
- Average of practical assignment grades needs to be a 5.5 or above
- Minimum grade for the exam is a 5.5
- Average of both assignments has to be 6.0

Literature and Study Materials

- Slides (released shortly before lectures)
- Scientific papers
- Lecture recordings

Prerequisites

- Human-Robot Interaction (Q2 course)
- Dynamics and Control (Q1 course)
- Programming (Python)

Assessment

Summative assessment:
- Exam
- Reports on practical assignments and implementation analysis
- Presentation of the final project

Formative assessment:
- Homework assignments

Exam:
The exam will be a computer exam in ANS on campus. In the case of COVID-19 restrictions, the exam will be held online (still in ANS).

Grade calculation:
-If and only if the exam grade is 5.5 or higher and the average assignment grade is 5.5 or higher, the grade will consist of 60% of the exam grade and 40% of the average score on the assignments. The final grade must be 6.0 or higher to pass the course.

Assignment submission lateness criteria:
<1 hour: -1 point
<1 day: -2 points
<2 days: -4 points
<3 days: -6 points
>3 days: no points given

Enrolment / Application

Registration:
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Department

ME Department Cognitive Robotics

RO47014	Knowledge Representation & Symbolic Reasoning	5
Responsible Instructor	C. Hernandez Corbato	
Contact Hours / Week x/x/x/x	0.0.6.0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an overview of knowledge representation and reasoning (KRR) techniques and their practical application in robotics, to support the programming of the robots behaviour with reasoning using symbolic knowledge about the mission tasks, its capabilities and its environment. The course will use existing programming frameworks such as ROS to practice using the KRR methods in realistic robotic systems. From the beginning, the focus will be on applying the theory to solve robotic problems, with practical assignments every week that will contribute to the final course project. Week 1: Introduction to Knowledge Representation and Reasoning Week 2: Ontologies Week 3: Graph-based methods Week 4: First-order, modal and multimodal logics Week 5: Planning with PDDL Week 6: Guest lecture Week 7, 8: Hands-on to complete the course project	
Study Goals	After completing this course, students will be able to: provide a comprehensive overview of Knowledge Representation and Reasoning methods, and list the main advantages and issues explain the differences between the most relevant logic-based representation languages for robotics, and their advantages and disadvantages model a robotic application in a logic-based language and use an off-the shelf reasoner to make inferences about the robot perceived environment select an appropriate representation and reasoner for the high-level control of a robotic application read related research literature and discuss with researchers in the field	
Education Method	Contact hours: Lectures, including in-class exercises (2 hours per week) Practicals (2-4 hours per week) to learn by doing with the support of the course staff. Course project The students will develop a KRR-based robotic application in groups of 3, using the techniques learned through the course. The application consists of a mobile manipulator in a simplified RoboCup@Home scenario.	
Literature and Study Materials	Lecture slides (released shortly before lectures) Text Book: Russell, Stuart J., and Peter Norvig. Artificial intelligence: a modern approach. Pearson Education Limited, 2016. Scientific papers	
Prerequisites	RO47003 Robot Software Practicals (needed) Programming in Python (needed) and C++ (recommended) Linear algebra, set theory Recomended: RO47005 Planning and decision making Recommended introductory course on logic (e.g. CSE1300 Reasoning and Logic) Recommended introductory course AI (e.g. IN4010-12 Artificial Intelligence Techniques)	
Assessment	5 written assignments, individual (40%), min. average of 5.0 Course project in groups (60%), min. average of 5.0 The students will give a presentation of their project results to an audience of robotic experts, submit a written report in the format of a scientific paper and upload the code to their course repository. All three elements are used to assess the students' projects.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Artificial intelligence	
Department	ME Department Cognitive Robotics	

RO47015	Applied Experimental Methods	5
Responsible Instructor	Dr. Y. Vardar	
Instructor	Prof.dr.ir. D.A. Abbink	
Contact Hours / Week x/x/x/x	0.0.0.x	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Study Goals	After completing this course, students will be able to: independently design, execute and analyse a human factors experiment formulate a clear hypothesis in the field of human factors that can be tested experimentally, in a realistic amount of time with a realistic amount of means set up a methodologically sound experiment to test this hypothesis o select, adapt or build an experimental setup o gather reliable experimental data on test subjects o choose appropriate metrics o apply correct statistical techniques o choose the most relevant results, and construct the most clear figures to show them critically reflect on your own experimental findings communicate clearly through presentations and writing	
Education Method	Students will perform a mini-graduation project in six weeks, in small groups (2-3 students). They will independently design, perform, analyze and present a human factors experiment, guided by four supervisors.	
Literature and Study Materials	Scientific papers	
Assessment	The final grade is composed of the following sub grades 10% Presentation 1: Hypothesis 10% Presentation 2: Methods 10% Presentation 3: Analyzing Results (two main results figures, explain) 20% First Draft for Paper 50% Final Paper In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment can take place in a similar way.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Broad Design Group Dynamics/Project Organisation Group work Integrated Intensive Involved Lab Research Project planning / management Small groups	
Department	ME Department Cognitive Robotics	

RO47017	Vehicle Dynamics and Control	5
Responsible Instructor	B. Shyrokau	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course provides fundamental knowledge on vehicle motion, vehicle-road interaction and the main technical principles and aspects of automotive control systems. This course discusses vehicle modelling, controller design and testing procedures to evaluate vehicle motion and automotive control systems. Tire dynamics and modelling Wheel slip, slip angle, tire force / moments, influence of camber angle, pressure, temperature effect, combined slip, friction circle, transient behavior Tire modelling and wheel dynamics Lateral motion and vehicle handling Kinematics, understeer gradient, bicycle model, steady-state handling Transient motion, frequency response, handling testing, vehicle stability Antilock braking system (ABS) and vehicle dynamics control (VDC) General concepts of ABS and different control principals: rule-based, mixed slip-deceleration, advanced control techniques General concept of VDC, yaw rate based controller, body slip angle based controller, integrated vehicle control using control allocation Planar motion Planar vehicle modelling Static equilibrium, lateral load transfer, roll center and axis, roll stiffness, anti-roll bars Longitudinal load transfer, pitch center, anti-dive, anti-lift Steering & assist logic Wheel kinematics, steering system modelling, column- and rack-assist Steering assist logic, steering feel and assessment Path-following and evasive control of automated vehicles Vehicle road model, error definition, an overview of path-following controllers, PD controller Evasive maneuver, advanced control techniques, MPC-based evasive steering assist Vertical motion and comfort Introduction to vibration, quarter car, 2DoF and 3DoF models, Ride comfort Suspension control: skyhook, groundhook, and mixed control Vehicle state estimation General concept of VSE, vehicle velocity estimation, Kalman filtering (linear and extended), wheel load reconstruction methods Final presentations of the group project	
Study Goals	After successfully completing this course, you will be able to: explain the tire-road interaction and be able to implement the tire model and compare it with the established one describe the concept of vehicle handling and lateral stability and derive the equations of motion of a nonlinear vehicle model interpret the load transfer effects and implement four-wheel planar vehicle model including the wheel kinematics, steering system and steering assist logic explain the concepts of wheel slip control, vehicle stability control, automated lateral control and vehicle state estimation implement control strategies for vehicle stability control (PID with gain scheduling or LQR) and automated lateral control (MPC) derive the equations of vertical motion of a quarter-car vehicle model and explain the influence of vehicle and road parameters on vehicle vibration behaviour	
Education Method	Lectures (4 hours per week) Office hours (2 hours per week) Group project & presentation	
Prerequisites	Basic programming experience (e.g. Matlab / Simulink) at the BSc level Dynamics and control at the BSc level knowledge about Model Predictive Control is recommended	
Assessment	Homework assignments (25%) individual Group project (25%) collaboration in groups of 4-5 students Written exam (50%) individual The examination is a closed-book digital exam on campus including multi-select and open questions. The grade for the homework assignments will be the average grade of the 3 assignments. Knock-out criteria o Minimum average grade for homework assignments is a 5.0 o Minimum average grade for group project is a 5.0 o Minimum grade for the exam is a 5.0 o Final grade minimum is a 5.75	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the	

	OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics
Contact	Barys Shyrokau, b.shyrokau@tudelft.nl

RO47018	**Security and Privacy in Control**	**3**
Responsible Instructor	L. Ferranti	
Contact Hours / Week x/x/x/x	X/X/X/4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Control Theory. Model predictive control. Convex optimization. Fault-tolerant control.	
Course Contents	Mobile robots, such as autonomous cars, vessels, and drones, can significantly improve our quality of life. These robots are equipped with communication units that allow information sharing and coordination near humans, abilities that are fundamental to improve, for example, traffic efficiency and even save human lives. However, the occurrence of attacks can severely disrupt the coordination with negative consequences on humans and their security. Furthermore, to coordinate, the robots share private information, such as habits, routes, and destinations, which may be inadvertently exposed to prying eyes. Hence, there is an urgent need to address these security and privacy concerns for our society to fully benefit from multi-robot systems. The goal of this course is to provide the students with (i) an overview of the state of the art in security and privacy from the perspective of control and robotics applications, and (ii) the basics to use state-of-the-art tools to design secure and privacy-preserving control algorithms. The course spans over the following main topics: 1) differential privacy and its applications to control. 2) applied cryptography (basics and more advanced concepts, such as secret sharing, garbled circuits, and secure multiparty computation) with some examples of applications to control systems. 3) an overview on the more general security aspects for control systems (discussion on the challenges and some applications based on recent research in the CS and control communities).	
Study Goals	The student is able to: o Understand the notions of security and privacy for control applications. o Critically reflect on the state of the art in security and privacy for control applications. o Reproduce and design secure and privacy-preserving algorithms for control applications	
Education Method	1. recorder video lectures, with discussions in class. 2. Supplementary material to gain the basic control and optimization background provided, for example, in the form of extra reading material, videos. 3. The format of the class will partially follow a seminar structure to stimulate discussion on state-of-the-art papers. 4. Invited lectures from experts in the field (if possible).	
Assessment	· Written review of a paper of choice and discussion of the paper with their peers during a debate lecture [1 EC, 40% of the final grade, mandatory to complete the course]; · Practical assignments in the form of short written reports that contain a reflection on the results with plots supporting the statements and accompanied by the code used to solve the assignments (e.g., using Python, SPDZ, and MATLAB) [1 EC, 40% of the final grade, mandatory to complete the course]; · Final exam (either oral or written depending on the number of students) [1 EC, 20% of the final grade, mandatory to complete the course].	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Cognitive Robotics	

RO47020	Advanced Machine Perception	5
Responsible Instructor	Dr. H. Caesar	
Instructor	Prof.dr. D. Gavrila	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course does a deep dive into Advanced Machine Perception techniques in robotics, with a focus on intelligent vehicles. It builds upon the Machine Perception course (RO47004) and teaches general Deep Learning concepts, as well as their application to 2D and 3D object detection. Introduction 1. Motivation: Deep Learning for Machine Perception Content overview Organization 2. Methods: Multilayer Perceptrons (MLPs) Recap basics (cf. Machine Perception, RO47004) Improvements (e.g. dropout, batch normalization, weight decay, adaptive learning) 3. Methods: Convolutional Neural Networks (CNNs) Recap basics (cf. Machine Perception, RO47004) CNN-based architectures and distinguishing features (e.g. LeNet (weight decay), AlexNet (deeper, ReLU, dropout), VGG (blocks), ResNet (skip connection), DenseNet (dense forward connections) Neural Architecture Search and EfficientNet 4. Applications: Unimodal object detection Single- and two-stage 2D detectors Image- and point cloud-based 3D detectors 5. Applications: Multimodal object detection Early/mid/late fusion Fusion architectures 6. Methods: Context and attention RNNs & LSTMs Attention & Transformers The lectures are subject to (minor) modifications with respect to content and quantity. The course has a substantial practical component (practicum and final assignment), where learned concepts are put into practice by means of programming assignments (PyTorch, using Jupyter notebooks).	
Study Goals	By the end of this course, students should be able to: [LO1] Understand the fundamental concepts of Deep Learning (e.g. CNNs and transformers). [LO2] Apply these concepts in practice (i.e. programming in PyTorch). [LO3] Train and evaluate multimodal 3D object detectors on a real-world dataset.	
Education Method	- Lectures (2 hours per week, for 6 weeks) - Practicum (2 hours per week, for 3 weeks) - Project (in groups) Formative feedback: Practicum - Pairwise collaboration (random assignment) - Pass/Fail (counts for knock-out criteria) - No late submissions or resit (except due to motivated medical conditions) Students can in principle run the software locally on their own computers (laptop/PC). However, we provide no support for setting up the necessary software environment (e.g. Anaconda). If the software environment cannot be installed properly (e.g. a few processors like M1 and ARM might not work), the student will need to revert to using the computers in the designated lab rooms. For tasks that require a Graphical Processing Unit (GPU), we will make a suitable online platform available.	
Assessment	Final assignment (60%) - Group project (random assignment) - Late submission: -1 grade point per day - No resit Students having an initial grade of < 5.8 will be offered an opportunity for a remake to obtain a (maximum) grade of 5.8 for this component in the following quarter. Exam (40%) - Individual - Written closed-book exam - Exam in Q4, resit in Q1 Knock-out criteria - At most one of the Pass/Fail practicum assignments is a Fail - Minimum grade for final assignment is a 5 (of current or previous academic year) - Minimum grade for the exam is a 5 - Final grade minimum is a 6 The final practicum assignment will be evaluated on the computers in the designated lab rooms. If a student chooses to work using their own computing equipment, the student needs to test the software on the designated computers before handing-in. Both final practicum assignment and final written exam results remain valid for one year. Each result can separately be transferred to the next academic year, upon request.	
Elective	Yes	
Tags	Algorithmics Artificial intelligence Image processing Programming Software Transport & Logistics	

RO Recommended elective courses on Transferable skills (between 3 and 6 ECTS)

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one’s own horizon and an (inter-)active scrutiny of one’s social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	

Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course.
	
	The selection procedure for this course will be as follows:
	- The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows:
	  - Q2 between September 30th at 08:00 and October 20th at 23:59.
	  - Q4 between February 3rd at 08:00 and February 16th at 23:59.
	Applications submitted outside of these slots will not be considered.
	
	- In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course?
	- The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received.
	- The results of the selection process will be communicated via email within ten working days after the closing of the application period.
	- The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4321-15	Air Traffic Management	4

Responsible Instructor	Prof.dr.ir. J.M. Hoekstra
Instructor	Dr.ir. J. Ellerbroek
Contact Hours / Week x/x/x/x	0/2/2/0
Education Period	2 3
Start Education	2
Exam Period	3 4
Course Language	English
Course Contents	The course gives an Introduction in Air Traffic Management and ATM research topics. 1. Introduction and history of Air Traffic Management, organisational & regulatory aspects 2. Airspace set-up, flight planning and navigation 3. Communication: Information services, Voice R/T, Data Link 4. Navigation: Great Circle calculations, WGS'84, approximations, beacons/navaids, landing aids 5. Surveillance: Radar basics, primary-secondary radar, radar types, ADS-B 6. Separation: Separation Minima, Near Misses, Closest Point of Approach Calculations, wind vortex categories 7. New concepts: Airborne separation assurance, 4D contract TBO, paired approaches, etc. 8. ATC Simulations: fast-time, human in the loop 9. ATC Research organisations and large scale programs (SESAR, NextGen) 10. ATM research topics: interactively discussed in class based on presentations by students While the first part of the course is mainly lectures, the second part of the course is much more interactive. Here pairs of students get assigned a paper, project or case to study and form an opinion on. They should explain and review it in both a brief essay and during a short presentation in class, followed by a discussion with the class in which they need to defend their view on this study. In this way the students are familiarized with different aspects and opinions on the 'hot topics' in ATM research currently.
Study Goals	This course is designed to give the students an appreciation of the current ATM systems, throughout the world, with emphasis on the core area of Europe, in preparing the existing ATC systems for the predicted increase of traffic. Emerging technologies will be described, with the associated implementation aspects like Air Traffic Controllers' systems, the history and future of ATC, airline perspective, European and US new ATM programmes SESAR and NEXTGEN. Emphasis is on both the current technology as well as future concepts and technology which are currently being researched.
Education Method	Lectures will be given on-line via livestreaming. Assignments will be given to the students for the second period. Students need to present their understanding of and opinion on a case, project or paper on-line. Discussions on this with the other students are part of the interactive method used in the second period.
Reader	Hand-outs of the presentations are available on Brightspace.
Assessment	Written, on site exam (50%) and the assignment (essay+presentation, also 50%). Both parts need at least a 5.0.

EPA131A	Behaviour in Transitions	5
Module Manager	Dr. G. de Vries	
Instructor	G. Scholz	
Co-responsible for assignments	G. Scholz	
Contact Hours / Week x/x/x/x	0/0/4/0 Only for students in the EPA MSc programma	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Only for students in the EPA MSc Programme	
Course Contents	Transitions in society are very complex. Their trajectories are uncertain and require behaviour change from all actors involved. For example, in energy transitions, policymakers must design policies that push for technological, system, and social innovations, and citizens and organisations need to accept and use these innovations. For informed policy analysis and policymaking, it is necessary to formalise and model this multi-actor behaviour. However, knowledge about its underlying drivers and barriers is needed to model behaviour. This knowledge can be provided by the scientific field of psychology and other areas, including economics, sociology, and philosophy. In this course, you will learn how behavioural insights (from various scientific disciplines) can help facilitate the formalisation and modelling of behaviour in transitions. More specifically, you will learn how behavioural perspectives, theories, and research methods can be applied to analyse the scientific understanding of multi-actor behaviour and the effectiveness of models. Application is challenging because human behaviour varies widely due to individualistic differences, biases, context and social interactions. Simply said, it is hard to translate behaviour into ones and zeroes. In this course, we take on this challenge.	
Study Goals	After taking this course, students can: Identify and describe behaviour from various actors in societal issues. Explain, discuss, and compare theories on individual and group behaviour. Discuss the usefulness of behavioural theories and behavioural data for simulation models and policies. Investigate and identify the contextual, social, and psychological factors that affect behaviour in transitions. Analyse the impact of contextual, social, and psychological factors on behaviour in transitions.	
Education Method	On campus lectures Workshops Q&A sessions Self-study of the literature on Brightspace	
Assessment	The assessment of this course consists of two parts. Both grades make up 50% of the final grade and are registered separately in Osiris. Both parts must be graded with a minimum of 5.5. Once registered, the results are transferable to the next academic year. The average final grade is at least 5.75. 1. An individual digital exam - existing of open questions in ANS - assesses the acquired knowledge on behaviour in transitions. Model exam questions plus answers - will be provided at the start of the course. There is a resit possibility in Q4. 2. A group assignment in pairs results in a written report on a behavioural analysis. Students work on this assignment throughout the course during workshops. A detailed description of the assignment, plus the assessment form, will be provided at the start of the course. When a pair receives a grade lower than 5.5., they get one repair option to a maximum of 6. If they fail again (<5.5), they can hand in a new assignment - on a new topic - in Q4.	

EPA143A	Macro-economics for Policy Analysis	5
Module Manager	Dr. S.T.H. Storm	
Co-responsible for assignments	Dr. C.W.M. Naastepad	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	None	
Summary	Economics is about how we work, what we produce, how we interact, cooperate and compete, and how we distribute and ultimately use what we have produced. Because the economy is too important to be left to the economists, this course provides an introduction to economics that explains how a market economy works and develops, why it often does not work as well as we would like, how its macro-economic performance can be improved by regulation and (fiscal, monetary or other public) policy. Particular attention is given to conflicts between achieving different goals of macroeconomic policy at the same time and to different theoretical (modelling) approaches. The course reviews: the neoclassical macro model; the (post-)Keynesian macro model; the IS-LM model; the multisector input-output model; the model of a balance-of-payments constrained developing economy; and a model of a technologically dual economy. The course does not assume that the student has any prior knowledge of economics.	
Course Contents	To assess the macroeconomic impacts of policy economists use models of how the economy is supposed to work and how major actors (consumers; producers; employees; government) are supposed to take decisions. Such models are always based on prior theoretical assumptions. This course reviews the major macro-economic theories as well as their operationalization in terms of (applied) macro and CGE models. Theoretical approaches reviewed include: Neoclassical; Keynesian; input-output modelling; IS-LM models. We assess the model assumptions, mechanisms and the policy conclusions. Keywords: Macroeconomics: aggregate demand and aggregate supply; circular flow of incomes; national accounting; neo-classical and Keynesian economics; unemployment; fiscal and monetary policy; money creation; crowding out; stabilization policies; labour market regulation; financial markets; balance-of-payments constrained growth; input-output analysis.	
Study Goals	At the end of this course, students will be able to: 1. analyze and weigh up the short-term and long-term impacts of macroeconomic policy on income, employment, productivity and competitiveness. 2. critically examine and reflect on the existing alternative macroeconomic theories, models and policy debates. 3. understand how money is created and how the financial sector works, and how this affects the real economy. 4. assess the strengths and limitations of an applied macroeconomic model analysis of a real-life policy issue. 5. productively work with an input-output model (of the world economy).	
Education Method	Lectures. Students are expected to prepare for each class by reading/studying the indicated literature.	
Literature and Study Materials	To be announced.	
Assessment	The exam is a closed-book electronic (in-person, on campus) examination of 3 hours. The on-campus exam will be made in Ans. The passing grade for the exam is 5.8. Students will be offered two opportunities to do the exam in an academic year. In case of unforeseen circumstances (such as due to COVID-19), the exam will be given in the form of an on-line (remote) open-book & timed multiple-choice examination of 3 hours. Students may review their exam answers. The exam review will be held physically on campus (and on-paper) according to the time scheduled by the lecturer(s).	
Special Information	None	
Elective	Yes	

MT44070	Shipping Management	5
Responsible Instructor	Prof.dr.ir. E.B.H.J. van Hassel	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course offers a comprehensive insight into the transport economic aspects of a maritime transport chain. This includes: - Short history of maritime trade and transport - Fundamental of transport economics - Maritime (sub) markets - Risk and Risk Management in shipping - Ports - Inland navigation - Hinterland transportation - Emission reductions (legislation and current developments)	
Study Goals	The objective of this course is to provide the students the insights into how ships are used to earn money. In order to do so, the functioning of the maritime markets needs to be understood. Also, it is the aim to explain that ships are part of a larger logistics chain and that the interaction in the port and with the other transport modes is vital. Next to that, also the working principles of a port are explained even as hinterland transport. Finally, also the current developments regarding emission reductions in the total transport chain are dealt with.	
Education Method	Physical lectures with audiovisual presentations.	
Assessment	Final exam in written form (75%). Group coursework (2 to 3 students): a short paper based on an assignment for which a model will be provided to the students (25%).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: MTM313-15	
Department	ME Department Maritime & Transport Technology	

TPM024A	Methods for Risk Analysis and Management	5
Module Manager	Dr. M. Yang	
Instructor	Dr. M. Yang	
Responsible for assignments	Dr. M. Yang	
Co-responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course consists of three blocks following a risk analytical approach: (A) introduction, (B) analysis, (C) assessment, treatment and follow-up. In block A Students will get familiar with the concepts and definition used in safety science and the basic steps of a risk analysis. In block B the course will focus on identification of hazards and qualitative/semi quantitative risk analysis. A selection of methods is explored to get acquainted with its application and limitations. For instance: Fault tree analysis, Event tree analysis, Bowtie diagram, Reliability Block Diagram, Markov Chain Analysis, Bayesian networks, Decision making under uncertainty. In block C the assessment of the outcomes of a risk analysis are put central. This includes acceptance, stakeholder perspectives, decision making and monitoring.	
Study Goals	After taking this course the student will have knowledge about: * the basic concepts of safety science and risk analysis * risk acceptance and decision making about risks After following this course, the student will be able to: * apply structured methods for risk identification, risk assessment, risk evaluation, and risk treatment.	
Education Method	The following methods may be used: Lectures, exercises and workgroup assignments. Students are expected to form groups to work on the projects applying risk-based methods in design or operation of socio-technical systems.	
Literature and Study Materials	The following textbook will be partially used: Engineering Risk Management (Meyer and Reniers, 2016; De Gruyter Publishing) Other required and additional material will be distributed via Brightspace	
Assessment	The primary assessment item of the course is a group project with an expected group size of 5-6 (depends on the class size). Assessment will be carried out on basis of groups at different stages of the project period. It will consist of three assignments, one presentation, and one final report. In total, there are 5 summative assessments. The grade distribution is: assignment 1 (proposal) - 15%, assignment 2 (methodology) -15%, assignment 3 (case study and results) - 15%, assignment 4 (oral presentation) - 25%, and assignment 5 (final report) - 30%. The minimum grade for each assessment item is 6,0. Feedback on the graded assignments can be reviewed by students upon appointment with the teach staff of this course within a week after the release of grades. The students are allowed one re-sit per assignment. It is not allowed to request a re-sit for an assignment for which students have received a pass (6,0 or higher). The re-sit format needs to be discussed with the teaching staff in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit must be taken within the current study year.	
Elective	Yes	
Category	MSc level	

TPM027A	Cyber Risk Management	5
Module Manager	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Dr. A.P. Afghari	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	MOTIVATION: The challenge of selecting the optimal technical, organisational, legal, and other preventive and repressive measures to reduce cyber risks to acceptable levels can only be understood in the context of the application of Cyber Risk Management. Risk Management is about analysing the relationships between threats, incidents and risks (here in the complex world of cyberspace), based on which an adequate set of countermeasures can be designed. SYNOPSIS: Risk (= the potential to losing something of value) can manifest itself in cyberspace in all kinds of ways: values at stake are financial wealth, health, physical condition (of people, materials, goods, infrastructures, etc.), well-being, reputation, privacy, trust, etc. Based on a conceptualisation of cyberspace and its various sub-domains (discussed in the project week of year 1), we introduce risk assessment approaches, both of qualitative and quantitative manner, illustrated with case studies. CONTENT: Cyberspace Risk concepts Probabilistic and adversarial risk System and attacker models Requirements and policies Risk assessment methods Qualitative and quantitative methods Measuring security and effectiveness	
Study Goals	After this course, students are able to: Understand and apply risk concepts to (possible) incidents in cyberspace Understand the theoretical principles of cyber risk management Explain the role of adversarial risk models, system models, and attacker profiles in cyber risk assessment Analyze and apply state-of-the-art cyber risk assessment methodologies to (complex, multi-step) cyber incidents Analyze the (expected) effect of measures that help to prevent the occurrence of cyber incidents Justify investments in cyber security	
Education Method	LECTURES: There are two lecture slots per week EXAM: There is a written exam at the end of the course. LANGUAGE: The course is taught in English. LECTURERS: Prof. dr. ir. Pieter v. Gelder (TUD/TPM) and guest lecturers.	
Assessment	#### Assessment Grading will be based on the individual written exam at the end of the course. Final mark of this course is determined by the grade of the individual exam (100%). The minimum grade for the individual exam to pass this course is 6.0. The grade for the individual exam is 2 years valid. The individual exam is summative. Detailed feedback on the individual exam can be reviewed by students by contacting the teaching staff of this course. #### Resit The students are allowed one re-sit per examination. It is not allowed to re-sit an examination or assignment for which they have received a pass (6,0 or higher). It is allowed to re-sit an examination which they haven't done during the first occasion. The re-sit format needs to be discussed with the teacher of the course in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit has to take place within the study year 2023-2024.	
Elective	Yes	
Targetgroup	MSc	
Category	MSc level	

TPM033A	Sustainable Innovation and Transitions	3
Module Manager	Dr.ir. J.N. Quist	
Instructor	Dr. L.M. Kamp	
Co-responsible for assignments	Dr. L.M. Kamp	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	TPM specialisation on Emerging Technology-based Innovation and Entrepreneurship	
Course Contents	The course Sustainable Innovation and Transitions takes as starting points: (1) that technology and innovation on the one hand and society and users on the other hand mutually influence one another and evolve in a co-evolutionary way; (2) that sustainable innovations, as well as system innovations and transitions towards sustainability, are strongly needed to bring about sustainability, and; (3) that stakeholder involvement and participatory intervention instruments are needed and required to enable and realise implementation in a socially responsible way. The course evolves around the concepts of Functions of Innovation Systems, Strategic Niche Management, Socio-Technical Transitions, social innovation and sustainable business models. These frameworks and instruments can be applied to sustainable or social innovations, which can be a technical innovation (e.g. recycling, EVs, biogas), an organisational innovation (e.g. car sharing), or a bottom-up grassroots innovation (e.g. energy cooperatives, urban agriculture) in both industrialised and developing countries. In addition, intervention instruments like transition management and backcasting are part of the course. The course consists of two parts: (1) A theoretical part consisting of lectures, readings and discussions, which is assessed by a written exam. (2) A group assignment focusing on a self-chosen topic. Examples of systems that can be studied include regional eco-industrial parks, organic agriculture, circular economy systems and renewable energy innovation systems in both industrialised and developing countries.	
Study Goals	By the end of the course students have: A. Knowledge of (i) major system innovation theories & transition theory; (ii) frameworks of Functions of Innovation Systems, Social Innovation, Strategic Niche Management; (iii) main concepts for sustainable business models; (iv) Major multi-actor intervention instruments at the level of socio-technical systems, such as transition management, and backcasting. B. Understanding of: strengths and limitations of the above mentioned theories, frameworks, tools and instruments. C. Experience & skills in the application of the frameworks to a self-chosen topic in a step-by-step group assignment. Academic skills that are trained in the course include (i) development of research questions & proposal; (ii) writing coherent, well-structured and well argued academic reports; (iii) dealing with user and social aspects of innovation, and (iv) stakeholder analysis and plurality in stakeholder perspectives.	
Education Method	Lectures, tutorials, group work on innovation analysis, presentations, individual assignment,	
Literature and Study Materials	Readings consist of selected journal papers and book chapters on the topics relevant to the course; a reading list of articles and chapters will be provided.	
Assessment	Summative assessment consists of: (1) an individual open book written exam using open questions (that counts for 50%), (2) a group project on a chosen sustainable or social innovation (also counting for 50%). (3) A bonus assignment of 0.8 point on the exam grade is possible (till a maximum exam grade of 10.0), consisting of weekly reading assignments and the peer review of draft work of another group. Formative assessment takes place through peer review of draft work of another group as part of the group assignment. The final grade is calculated as follows: (1*written exam + 1*bonus assignment+1*group project)/2. Final grades are expressed by means of a figure between 1 and 10, rounded to the nearest half. Both exam (complemented with the bonus assignment point) and group project must be completed with at least 5.0 and the final grade must be at least or higher than 5.75. Only the final grade will be entered in OSIRIS. Please register for the exam in OSIRIS on time (see: https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams).	

TPM0637SET	Economic Policy for Sustainable Energy	4
Module Manager	Dr. E. Schröder	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge of algebra and calculus is required. In general, the typical master student in a STEM subject has mathematical competencies that exceed those required for this course. The course builds on some of the ideas covered in the SET course Economics and Regulations of Sustainable Energy Systems in Q1, especially the parts taught by Gideon Ndubuisi and Servaas Storm. We will review some of these ideas, but in general non-SET students will have to catch up.	
Course Contents	Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis? This is a course in environmental and ecological economics designed for master students with little background in economics. It introduces students to key economic concepts and theories with respect to: - consumption and production - efficiency and welfare - economic growth and sustainability - the choice of environmental policy targets and instruments - integrated assessment models	
Study Goals	After completing the course, students should be able to: - explain the main theoretical concepts used in environmental economics and how they relate to one another - explain how a market economy undervalues the environment - explain how environmental economists value the environment - apply economic theory to environmental problems such as climate change - evaluate economic policies for solving environmental problems such as climate change	
Education Method	Classes, quizzes, and exercises.	
Literature and Study Materials	The main text is: - Harris & Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor & Francis. The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.	
Assessment	The final grade is based exclusively on the performance in the final exam. In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.	

TPM411A	Idea to Start-up IT & AI	5
Module Manager	E. Çelik	
Instructor	E. Çelik	
Instructor	Dr. A. Giga	
Responsible for assignments	E. Çelik	
Co-responsible for assignments	Dr. A. Giga	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity using emerging digital technologies. Applications can range from mobility, cyber security to predictive maintenance etc. You will test its validity, and build a viable business model a roadmap from an idea to an innovation-driven company. On this journey, you will be coached by lecturers and learn practical insights from guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one.	
Course Contents	You will focus on information and computing technologies (e.g AI, big data, blockchain) that are transforming traditional industries and societies at large. To analyse the opportunities for new startups we focus on entrepreneurial tools such as the opportunity canvas; value proposition development and customer empathy mapping, market segmentation and value chain analyses and the initial operations to develop a minimum viable product.	
Study Goals	Develop a business Idea by identifying a relevant economic problem or unmet need that is addressed by digital solutions, and advanced information and computing technologies. Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment Analyse the competitive landscape and devise your products competitive advantage, given specific challenges of protecting IP in the digital world. Create a blueprint of your product. Identify relevant stakeholders in your startups ecosystem, outline the go-to-market strategy, and assess profitability. Understand how these strategies adjust when trying to implement your solution in well-established systems of traditional industries. Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by improving the business model. Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing	
Education Method	You will work in teams which are formed in the first sessions of the course. One aspect of a startup will be discussed each week and you will apply into your own projece immediately. You will present your work, provide peer-feedback as well and receive advise from coaches. The course ends with a jury pitch and a 4-pager of the business proposition. Team coaching will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Books	The primary textbook for this course is an organic handbook by Bill Aulet: Disciplined Entrepreneurship: 24 Steps to a Successful Startup. Wiley, 2023.	
Assessment	The assessment is based on your project. It includes class assignments and participation (40%), final business model report (30%), online pitch presentation (15%), and team evaluation (15%). Participation and Peer Evaluation grades pertain to each student. Class assignments, final business report, and pitch presentations are evaluated on a group level.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM416A	Turning Technology into Business	6
Module Manager	Dr. L. Hartmann	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Turning Technology into Business (TTiB) aims to equip participants with a strong conceptual foundation to actively understand the dynamic process of technology-based entrepreneurship. Participants learn how business strategies are best formulated and how (through entrepreneurship) technology can create value. TTiB consists of seven lectures, introducing the theoretical backgrounds of technological, market and business analyses. A unique aspect of TTiB is that existing technologies (developed and patented by the TU Delft) are used as case subjects. The patent project is the focal point of the curriculum. Each group of 4-5 students will be assigned an original patent, and is expected to evaluate the commercial potential of this technology. This includes choosing promising applications (products) for the technology, and making recommendations for the most suitable business model to commercialize the technology. The patent project should provide a coherent and structured answer to the central question: which strategy is most likely to generate business from this patent? In addition to equipping participants with a thorough knowledge of the course subject, the section Technology, Strategy & Entrepreneurship (TSE) hopes that this course will be the starting point for a variety of university spin-offs in which the students will participate. To accommodate this, TSE offers the course TPM420A- Ready to Startup!	
Study Goals	This course aims to equip students with a strong conceptual foundation to an active understanding of two domains: (1) The dynamic process of technological innovation through concepts such as technology life-cycles, dominant design, disruptive technologies, Schumpeterian competition and the diffusion of innovations (2) How business strategies are formulated and, through entrepreneurship, technology can create value. This multi-faceted process of technology commercialization process is addressed in terms of assessing technology position, discovering market opportunities, competitive analysis, appropriability and the various modes of entrepreneurship.	
Education Method	7 interactive lectures, participant-centered case studies, homework assignments, classroom assignments, individual group coaching. Final Presentations in February.	
Literature and Study Materials	Will be provided during the course	
Assessment	Each group (of 4 or 5 students) writes a report on their findings, analysis and recommendations for the technology (patent) they used as a case for this course. On the Final Day, the group has to give a 10 minute oral presentation. The report makes up 80% of the final grade. The final grade is summative. This team grade will be modified according to the individual student's class contributions and his/her performance within the group (peer review). There is no resit possibility.	
Enrolment / Application	MAXIMUM CAPACITY is limited to 75 participants. Register on BrightSpace from October 1. Because of the huge popularity of this course, there is a selection procedure which consists of pre-course assignments. One of these assignments is a letter of motivation: why do you want to participate in this course? Based on the motivation and the other pre-course assignments, the final selection of 75 participants will be made.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Advanced Master students, PDEng and PhD students, and staff researchers from all faculties within the TU Delft.	
Category	MSc level	

TPM420A	Ready to Start-up	6
Module Manager	P. Harish	
Instructor	Dr.ing. V.E. Scholten	
Instructor	P. Harish	
Responsible for assignments	P. Harish	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	This course is offered twice in a year; First starting week 3/4 of Q1 and ending in Q2 and second starting week 3/4 of Q3 until end Q4 Ready to Startup (RtS) is an elective master course that has been the crown jewel of entrepreneurial education at TU Delft for over two decades. Bringing together business experts and students eager to start companies, it has one simple goal: turn engineers into successful founders.	
Course Contents	RtS focuses predominantly on Business Strategy: gaining market traction, prototyping, growing the team, devising a financial plan and fundraising strategy, protecting IP, and foundational activities (setting up a company, sharing ownership, etc.) The course has the prerequisite that teams should be formed and have a business ideas that they want to further analyse and validate. It is meant for students who plan to start a company. After RtS, teams must be at the level required for YES!Delft incubation and Delft Enterprises investments.	
Study Goals	At the end of the course, you will be able to - Reflect on your current business model and identify further necessary validation activities. Analyse market entry strategies and consider alternative models. - Learn how to negotiate, attract first users, and seek endorsement of stakeholders (such as investors and advisors). - Plan your core operational activities such as company foundation, IP protection, product development strategy and team development - Devise financial plans and appropriate fundraising strategies - Convey a convincing narrative of your business and be able to provide constructive feedback to other startup plans. - Demonstrate effective entrepreneurial behaviour by making swift decisions based on newly gathered information and adjusting your model accordingly.	
Education Method	The course is spread out over 12 in-person sessions on Wednesday afternoon (13h30-17h30), where the core elements of a startup plan are a presented. You work on assignments that are aimed at specific elements of the core startup plan, and you will by apply them to your startup project and present your findings during the course. In this way you make evident progress with your startup. The course combines state of the art business methods, translated to the needs of starting entrepreneurs, with consults from practitioners. Topics are discussed using case studies. You will present your progress, provide peer-feedback as well as receive advice from mentor experts. The course ends with a Final Day, consisting of a jury pitch and handing in your business plan.	
Literature and Study Materials	Handouts and literature are available trough in Brightspace.	
Prerequisites	Enthusiasm, entrepreneurial spirit and a good business idea.	
Assessment	The course has 4 grading components. - Business plan: final report (40%) - Financial plan (10%) - Final Day performance (30%) - Overall progress: homework and attendance (20%) In general, you are only allowed to miss one lecture (with a valid reason submitted beforehand). Furthermore, we will monitor your progress during the course, with an official assessment halfway through the course.	
Enrolment / Application	Application process you need to register in advance to the course. Teams of students with clear ideas about a business they want to start supported with evidence that it addresses a need, will have priority to enter. ==> You can directly enroll in Brightspace. ==> If you need to use the mytudelft app, you may see that you cannot register to this course. BUT that information is for the exam, which does not apply to this course, since there is no exam. Hence, you can register and enroll in Brightspace. ==> You may need approval from your exam committee, when you take this course within your elective space. But please register also in Brightspace and we can plan the course. ==> You can always take this course as an extra course, not part of your elective list, and it will be included in grade list.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

WI4425	Financial Markets Theory	6
Responsible Instructor	Dr. J.H.M. Anderluh	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Choices under Uncertainty, General Equilibrium Theory, Pareto Optimality, Expected Utility Theory, Risk Aversion Choices under risk : Portfolio Problem, Insurance Demand en Prudence, Stochastic Dominance, Mean Variance Analysis Portfolio frontier, Mutual Funds separation, Risk sharing and Pareto optimality Asset Markets, Intertemporal Consumption, the F. theory of asset pricing, CAPM CAPM, APT, Portfolio Choice Multi Period Markets Information and Financial Markets Uncertainty, Rationality and Heterogenity Financial Markets Microstructure Corporate Finance	
Study Goals	The main goal is to enable the student to quantify and in some cases optimise the relation between return and risk in a financial markets situation. The student will be able to build a mathematical model of the situation at hand with ingredients like equilibrium concepts, no-arbitrage concepts.	
Education Method	One block of 2 hours lectures every week during the 1st semester. A written exam afterwards.	
Books	Emilio Barucci, Financial Markets Theory	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WM0329TU	Ethics and Engineering	6
Module Manager	M. Sand	
Instructor	A.R. Gammon	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	4/0/0/0 or 0/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	The first part of this course completely coincides with the WM320TU course. The WM329TU course is the WM320TU course plus the writing of an essay. Please check the WM320TU Brightspace for all relevant information about lectures, work sessions, exam, and the essay. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. are able to write an opinion article on ethical or societal aspects of technology, according to academic standards of writing are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and 5 interactive work sessions concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	50%: grade for the participation in the course WM0320TU, which consists of a written exam (60%, individual grade), presentation, and active participation during the working group sessions (40%, same grade for all members of a presentation group). In order to pass the WM0320 part of the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. These partial grades are only valid for 1 academic year, meaning that if one or both are not passed during resit, both assessments need to be retaken the following academic year. 50%: grade of the essay, which is written individually (all steps in the essay writing procedure need to be followed). The grade of the essay (50%) conjoined with the grade of the WM0320TU part (50%) must be higher than 5.75. Only if the threshold is not met will it be possible to revise and resubmit the essay. It is not possible to retake the essay in another semester.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the number of workgroups. Please enrol as soon as possible and in any case before the start of the course via Brightspace.	
Remarks	The course is run twice each year, starting in quarters 1 and 3. The first part of this course (concluded with the written test) coincides with the course WM0320TU.	
Elective	Yes	
Category	MSc niveau	

WM0376TU	Ethics of Technological Risks	5
Module Manager	Prof.dr. S. Roeser	
Instructor	K. Maheshwari	
Responsible for assignments	Prof.dr. S. Roeser	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course investigates ethical aspects of technological risks. Debates about technological risks related to, for example, energy technologies, robotics and biotechnology frequently culminate in stalemates. This is due to the complexities and intricacies inherent to such debates as they involve scientific information and uncertainties, as well as ethical and emotional considerations. Conventional, quantitative approaches focus primarily on statistical information to risk. They do not explicitly incorporate important ethical considerations such as justice, fairness and autonomy. Emotions such as compassion, care and feelings of responsibility can draw attention to such ethical aspects of risky technologies. Taking emotions and ethical considerations seriously can lead to more fruitful deliberations between different stakeholders in which relevant concerns are taken seriously and are explicitly reflected upon. This course will study how approaches to ethical aspects of risk can lead to more morally responsible decision making and design of technological innovations.	
Study Goals	This course will provide students with an understanding of approaches to risk ethics and students will get experience with discussing and evaluating ethical aspects of risk.	
Education Method	Self-study, lectures and tutorials (in smaller groups, ca 25-35 students, depending on the total number of students). All students will hold one group presentation (ca 5 students) at one of the lectures. Attending lectures and tutorials and giving a presentation is obligatory.	
Assessment	On campus written exam (60% of the final grade) and essay (40% of the final grade). The exam is an individual assignment, the essay is written in pairs of students. Students need to have a pass (5.75 or higher) for each individual grade in order to pass the course. There is a resit of the written exam in the 4th quarter (date to be announced). In case students did not pass the essay initially, they have the opportunity to make further revisions within ca 2 weeks, but then the maximum grade is a 6, given the extra time. Partial grades are valid for the current and the following academic year. In case of unforeseen circumstances, eg related to COVID-19 measures, the on campus exam may be replaced with an online (eg take home) exam [There is also a 3 ECTS version of this course, i.e. WM0375TU; the only difference is that for the 3 ECTS version students do not write an essay.]	
Special Information	Contact person for additional questions: S.Roeser@tudelft.nl	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Philosophy	
Category	MSc level	

WM0801TU	Introduction to Safety Science	3
Module Manager	Dr. A.P. Afghari	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Dr. A.P. Afghari	
Co-responsible for assignments	Dr. E. Papadimitriou	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Students wishing to graduate with a final project in the area of safety supervised by the Safety Science and Security Group should have followed this course or one of the other introductory courses run by the group (WM0808TU, WM0821TU or WM0822TU)	
Expected prior knowledge	Technical academic BSc and common (scientific) sense	
Summary	System Safety Engineering focuses on development of a safety oriented pattern of thinking and a holistic approach. The tools that will be gained in this course will be helpful in recognizing, understanding, and analyzing hazards; and assessing risks in contemporary complex systems. Areas included are (1) Hazard analysis and safety assessment, (2) Human reliability assessment, and (3) Safety management.	
Course Contents	Introduction to system safety engineering Safety performance measurement Hazard analysis Failure modes and effects analysis Preliminary hazard analysis System safety assessment Reliability block diagram Fault & event trees Markov chain analysis Bayesian network Consequence assessment Risk ranking Human reliability assessment Decision making under uncertainty	
Study Goals	The students will learn how to perform safety assessment, reduce risk within acceptable levels, manage risk, improve system safety, and make risk-informed decisions to benefit the organization and the community.	
Education Method	Plenary lectures (mandatory) Exercises (carried out in small groups of ~2 students)	
Literature and Study Materials	Clifton A. Ericson, Hazard Analysis Techniques for System Safety, John Wiley and Sons; ISBN: 978-0471720195. Harold E. Ronald and Brian Moriarty. (1990). System Safety Engineering and Management, John Wiley and Sons, 2nd Edition.	
Prerequisites	Technical background preferred but certainly not mandatory	
Assessment	Summative assessment: A final exam consisting of open-ended and multiple-choice questions; The exam is based on book, papers, exercises and lecture slides; The final exam is individual and accounts for 100% of the final grade for the course; A re-sit exam is possible if needed. Formative assessment: In-class quizzes, discussions and games; Sample questions for final exam.	
Enrolment / Application	Enrol through Brightspace. Minor students are automatically enrolled into the course.	
Remarks	This course can be expanded to a System Safety/Reliability project	
Elective	Yes	
Category	MSc level	

WM1302TU	Ethics of Transportation	5
Module Manager	Dr. E. Ziliotti	
Responsible for assignments	Dr. E. Ziliotti	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course is identical to the WM1301 course, with the addition of a final essay. In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems can see these problems within the broader background of moral and political philosophy have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain. are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way can write a short academic essay of critical analysis of an ethical or societal issue related to transportation technology	
Education Method	A series of 6 lectures, active participation in some working groups, and the writing of an essay on ethical, social or legal aspects of technology to be written under the supervision of an instructor from the section Ethics of Technology at TBM	
Assessment	Written exam, assessment participation in working groups and the writing of an essay the ethical, social or legal aspect of technology to be written under the supervision of the course manager or an instructor from the section Ethics of Technology at TBM. In order to pass the course both the exam and the essay should be graded with at least 5.5 (and the average should be sufficient, that is higher than 5.75)	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Robotics**

RO General elective courses	
Introduction 1	Machine Perception involves the capability of a machine to interpret data provided by environment sensors analogously to the way that humans use inputs from their senses to relate to the world around them. Such capability is indispensable to a robot when navigating and interacting with a complex and dynamic environment.

AE4301	Automatic Flight Control System Design	3
Responsible Instructor	Dr. A. Jamshidnejad	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4303	
Expected prior knowledge	The following prior knowledge is required: AE3302 AE2204 (until 2012/2013) AE2235-I (from 2013/2014)	
Parts	1. Introduction & recapitulation of systems and control theory . Laplace transformation . Elementary closed loop systems . Transfer functions . Poles and zeros . First order systems . Second order systems . Pole placement for simple systems . Root locus method . Characteristic equation . Angle and magnitude conditions . State space formulation . Controllability, observability . Basic controllers: P,PI,PD,PID . Frequency response (Bode plots) 2. Introduction: Course use and arrangement a. Why automatic flight control systems? b. Function of the flight control system in civil aviation c. Recapitulation of theory on flight dynamics d. Review of the different frames of reference: wind, stability, and body. e. Non-linear equations of motion. f. Trim and linearization of the non-linear equations of motion. g. Linearized longitudinal dynamics using a statespace representation and the equivalent frequency domain form. 3. Performance and handling qualities a. Handling quality criteria b. The Control Anticipation Parameter (CAP) c. Gibsons Phase rate and Frequency criterion 4. Dynamic stability augmentation a. yaw dampers b. pitch dampers c. phugoid dampers 5. Static stability augmentation a. angle of attack feedback to improve static margin b. load factor feedback to improve manoeuvre margin c. sideslip feedback to improve directional static stability 6. Basic longitudinal autopilot modes a. pitch attitude hold mode b. altitude hold mode c. airspeed hold mode (using autothrottle) d. vertical speed 7. Basic lateral autopilot modes a. roll angle hold mode b. coordinated roll angle hold mode c. turn rate at constant altitude and speed d. heading angle hold mode 8. Longitudinal and lateral guidance modes a. glideslope hold mode b. automatic flare mode c. localizer hold mode d. VOR hold mode	
Course Contents	Classical control is still predominantly used in aerospace industry for the design and analysis of automatic flight control systems. Various existing control systems such as Stability Augmentation Systems (SAS), Control Augmentation Systems (CAS) and fly-bywire systems are reviewed in detail. The emphasis of the course lies in demonstrating, through application of classical frequency domain and state space techniques, how to design systems that fulfill the requirements imposed by the aviation authorities, with emphasis on understanding the benefits and limitations of such systems.	
Study Goals	After this course the student should be able to: - Understand dynamics of linear time invariant systems and design controllers for such system for the desired performance	

	- substantiate the function of a Flight Control System(FCS) in civil aviation. - apply the theory of flight dynamics and control to FCS design for civil aviation. - verify if a given FCS satisfies the handling qualities for civil aviation criteria. - design static and dynamic stability augmentation systems. - design all longitudinal and lateral autopilot modes for civil aviation.
Education Method	Lectures
Literature and Study Materials	Course material to support the exercises will be posted on the Brightspace. Recommended literature - Any standard textbook on systems and control theory (e.g., Modern Control Engineering by Ogata) - M.V. Cook, Principles in flight dynamics, Edward Arnold, London, 1997 ISBN 0-340-63200-3. - B.L. Stevens, F.L.Lewis, Aircraft control and simulation, Wiley, New York, 1992 ISBN 0471613975. - J. Roskam, Airplane flight dynamics and control Part II, , ISBN 1-8845885-18-7.
Assessment	Written closed-book examination
Set-up	At the end of each lecture, a simple take home assignment is given in order to gain experience in working with the course material. There will be a written examination at the end of the course. In the related practical AE4301P a control system must be designed that satisfies certain desired requirements.

AE4316	Aerospace Human-Machine Systems	4
Responsible Instructor	Dr.ir. M.M. van Paassen	
Instructor	Prof.dr.ir. M. Mulder	
Instructor	Dr.ir. C. Borst	
Instructor	Dr.ir. D.M. Pool	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Follow-up course (elective) AE4316P. That course focuses on advanced topics in aerospace human-machine systems by having small groups of students participate in an experiment (in either a manual or supervisory control setting), collect data and analyze their own generated data. In other words, experience the theory behind human-machine systems in practice!	
Expected prior knowledge	Control theory (stability and Bode plots); signals AE2235 Aerospace Signals, Systems & Control, AE4301 Automatic Flight Control System Design Stochastic signals; AE4304 Stochastic Aerospace Systems Flight instruments and procedures; AE4302 Avionics and Operations	
Course Contents	This course focuses on the various aspects of actual and future flight deck and air traffic control human-machine interfaces. It provides extensive theoretical as well as practical knowledge on the specific characteristics of human behavior, such as human control behavior (i.e., cybernetics), human perception (visual & haptic), human mental processing, cognitive factors, and human-automation interaction in manual and supervisory control tasks.	
Study Goals	Overall, the student will have a working knowledge of human operator (pilot) characteristics that are relevant for the design and evaluation of human-machine systems. Specific study goals: The student: * Is able to classify different types of human behaviour according to Rasmussen's rule-skill and knowledge taxonomy * Is able to predict performance and human behaviour in manual control tasks, using McRuer's cross-over theory, and is able to reason on the effect of task variables (motion, haptics, etc.) * Knows the physiology and characteristics of human sensory systems and actuation processes (visual, vestibular, propioceptive senses and neuromuscular system) as relevant for human behaviour, and is able to predict the implications of these properties for human perception and behaviour, and relate this to design choices for displays and manipulators. * can describe the pitfalls of automation (i.e., ironies of automation) and is familiar with various taxonomies on levels & stages of automation. * can analyse accident and incident reports, find latent and active errors and classify these with Rasmussen's SRK taxonomy and identify Reason's error shaping factor. The student understands the wider context of human error (Dekker's "new view"). * is familiar with workload and situation awareness, and knows which methods are used to measure these properties * understands the nature of human cognition, can distinguish between different views and models of cognition and knows when these are applicable * can describe the differences and similarities between interface design approaches that aim to support both system and human (control & cognitive) performances * is familiar with workload theory, knows different metrics for workload, and when these are applicable	
Education Method	Lectures with activating elements. Week arrangement: 1. Introduction + SRK 2. Crossover model I 3. Crossover model II 4. Crossover Model II 5. Visual systems (the eye and visual motion perception) 6. Vestibular system 7. Neuromuscular system 8. Human-Automation Interaction 9. Human Error 10. Cognition 11. Ecological Interface Design 12. Display Design Principles 13. Applications of HMS theory 14. Workload (self study module, no lecture) The course also includes a demonstration session in the human machine laboratory, in which students will encounter experimentation settings and experience manual control tasks.	
Assessment	Written exam. The exam typically consists of 6 questions, covering a selection of the course's topics.	

AE4317	Autonomous Flight of Micro Air Vehicles	4
Responsible Instructor	Dr.ir. C. de Wagter	
Instructor	Dr. G.C.H.E. de Croon	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Programming experience (any language) If the student has no previous background in C / C++, we expect the student to do an online tutorial before/at the beginning of the course. Desired prior knowledge: Programming experience with C / C++ AE4393 (Avionics and Operations)	
Course Contents	This course covers the challenges and existing state-of-the-art methods for enabling autonomous flight of Micro Air Vehicles (MAVs), ranging from 20-gram flapping wings to 1 kg quad rotors. The emphasis is on computationally efficient, bio-inspired approaches to MAV autonomous flight. The theoretical knowledge will be applied by the students in the practical assignment, in which student groups program quad rotors in order to avoid obstacles in TU Delft's Cyberzoo.	
Study Goals	(1) Students can explain the differences between MAVs and larger aircraft in terms of control, design, aerodynamics, and electronics. (2) Students can explain how the above-mentioned differences affect the task of achieving autonomous flight with MAVs. (3) Students can list and explain Artificial Intelligence (AI) and control techniques useful for autonomous flight of Micro Air Vehicles (MAVs). a) Bio-inspired robotics b) Vision-based navigation (4) Students can apply the learned AI and control techniques to, and create new algorithms for an autonomous flight task. (5) Students can reproduce the current state of regulations for MAV operations. In addition, they can assess and discuss the safety and ethical aspects of MAV operations.	
Education Method	The course will consist of theory (7 lectures) and a practical assignment (6 practical sessions and a competition session). Brief description of practical assignment: Student teams will develop an efficient, purely vision-based approach to navigate through an obstacle field autonomously. The algorithm must run onboard the Parrot Bebop Drone, a commercially available quadrotor with a range of sensors. Importantly for the project, it has a single frontal camera. Hence, the developed algorithm concerns a monocular vision solution to obstacle avoidance.	
Assessment	The assessment of the course will consist of 50% of the theoretical exam and 50% of the practical assignment. Assessment of the practical assignment is based on the performance of the drone in the given task, the approach as explained in a presentation, and the code produced by the team.	

BM41040	Neuromechanics & Motor Control	5
Responsible Instructor	Dr.ir. W. Mugge	
Instructor	Prof.dr. F.C.T. van der Helm	
Instructor	Prof.dr.ir. A.C. Schouten	
Instructor	Ir. M. van de van de Ruit	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Relevant TU Delft BSc Courses: -Wiskunde 2 Lineaire Algebra WBMT1051 -Wiskunde 3 Differentiaal Vergelijkingen WBMT2048 T2 -Wiskunde 4 Kansrekening & Statistiek and Numerieke Wiskunde WBMT2049 T1&T2 -Rigid Body Dynamics WB2630 -Signaalanalyse WB2235 -Systeem- en Regeltechniek WB3240	
Course Contents	Humans have an extraordinary flexibility in generating movements. With little attention most of us can perform complex tasks such as walking or riding a bicycle. The central nervous system integrates available sensory information and select the proper motor command for the muscles. The skeletal system enables the transfer of forces required for body movement while the muscles produce the forces to accelerate it. Stability is maintained through both passive (intrinsic muscle properties) and active (reflexes) contributions to the visco-elasticity of the joints. In this course we investigate how humans generate movements, stabilize posture, and integrate sensory feedback into control actions, while acknowledging the kinematics and the mechanical properties of the human motor system. This course addresses fundamental aspects and clinical applications. During the course the role of subsystems (e.g. muscles, sensory systems) and their interactions as well as uncertainty in sensory signals are discussed with a focus on reaching movements and posture maintenance. During the course it will be shown how this knowledge can help in understanding impaired motor function after neurological disorders, like Stroke and Parkinsons disease.	
Study Goals	After this course the student must be able to: - Describe the motion of bony segments of the human skeleton system with respect to each other using kinematic analysis methods. - Calculate muscle forces from known joint motion using different optimization techniques (forward dynamics and inverse optimization). - Describe the physiology of the human motor and sensory systems as well as the central nervous system with respect to motor control. - Build and use neuromusculoskeletal models to simulate movements using motion equations. - Explain the neuromuscular system from a control engineering perspective, including the interaction between intrinsic and reflexive feedback. - Describe how the central nervous system deals with uncertainty, integrates sensory feedback and adapts to new situations. - Is able to deduce how impairments of different processes involved in motor control due to neuromuscular disorders affect motor control and adaptation. - Is able to critically judge the value of theories about postural and movement control posed in scientific literature.	
Education Method	Lectures and practicals (4 hours per week) and homework assignments	
Assessment	Written/Digital exam, computer tests (digital midterm) and assignments	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Lab Research Matlab Mechanics Mechatronics Modelling Optimalisation Signals and Systems Technology	
Department	ME Department Biomechanical Engineering	
Contact	Preferably by email to the Responsible Instructor	

CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course covers the following topics: - basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and terminology of real-time systems 2. Construct task schedules using different scheduling policies under a given set of realistic system constraints 3. Analyze the timing behavior of a system for a given system model and scheduling policy 4. Discuss advantages and disadvantages of different scheduling policies for a given platform or system 5. Discuss the effect of hardware and software interferences on the timing behavior of a given system 6. Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system 7. Derive (reverse engineer) the system specification from a given implementation (in the lab) 8. Evaluate the scheduling overheads of a given implementation (in the lab) 9. Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Individual Assignment: lab work (weighting 40%) Final grade calculation = 0.6 * Written Exam + 0.4 * Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 6 (5.8 and higher). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4030	Embedded Systems Laboratory	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. G. Lan	
Instructor	Ir. J.B. Dönszelmann	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This highly multi-disciplinary course comes with a lab project where teams of 4 students each will have to develop an embedded control unit for a tethered electrical model quad rotor aerial vehicle (the Quadrupel drone), in order to provide stabilization such that it can hover and (ideally!) fly, with only limited user control (one joystick). The control algorithm (which is given) must be mapped onto a home-brew PCB holding a modern RF SoC interfacing a sensor module and the motor controllers. The students will be exposed to simple physics, signal processing, sensors (gyros, accelerometers), actuators (motors, servos), and basic control principles. The embedded software running on the flight controller board (FCB) driving the drone must be written in the Rust programming language. The project work (including written report) covers the entire duration of the course period, and will take approximately 128 hours, of which 32 hours are spent at the lab facilities.	
Study Goals	Student is acquainted with real-time programming in an embedded context, along with a basic understanding of embedded systems, real-time communication, sensor data processing, actuator control, control theory, and simulation. Moreover, the student has had exposure to integrating the various multidisciplinary aspects at the system level. At the end of this course, the student is able to: 1. Explain and analyse real-time communication, sensor data processing, actuator control, control theory in the context of an embedded system 2. Consider various multidisciplinary aspects at the system level 3. Apply real-time programming concepts in an embedded context	
Education Method	Lectures (6*2hrs), lab work (8*4hrs), coding at home (8*12hrs), report (8hrs), so on average 2 days per week	
Prerequisites	Literacy in Rust (CESE4000 or equivalent, and CSE4015)	
Assessment	The final grade of the course consists of the following components: - Lab Project (weighting 75%) - Written Report (weighting 25%) Final grade calculation = 0.75 * Lab Project + 0.25 * Written Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Lab Project: Additional assignment - Written Report: Additional report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for CESE students- they get preference over other MSc students.	

CS4205	Evolutionary Algorithms	5
Responsible Instructor	Prof.dr. P.A.N. Bosman	
Responsible Instructor	Dr.ir. P.A. Bouter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, (GPU) parallelization, and real-world applications.	
Study Goals	At the end of this course, the student is able to: 1. Explain the key concepts underlying the main streams in Evolutionary Algorithm (EA) research, with in particular genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2. Explain key ingredients underlying the rationale of when these algorithms work and when they do not work. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3. Explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4. Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5. Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	8-10 lectures 4 smaller (individual) projects 1 larger group project	
Literature and Study Materials	Various scientific papers, course-specific texts, and slides	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment (weighting 30%) - Individual Assignments: weekly assignments (weighting 10%) Final grade calculation = 0.6 * Written Exam + 0.3 * Group Assignments + 0.1 * Individual Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4235	Socio-Cognitive Engineering	5
Responsible Instructor	Prof.dr. M.A. Neerincx	
Instructor	Dr. B.J.W. Dudzik	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic prior knowledge on human-computer interaction is helpful, but not required.	
Course Contents	Artificial intelligence (AI) is increasingly integrated into various aspects of our daily lives, from work and leisure to care settings. Hybrid Intelligence (HI) aims to create a beneficial integration that leverages the strengths of human and artificial intelligence, while compensating for their respective weaknesses. By aligning AI with human values and cognitive processes, it can collaborate with humans to enhance hybrid decision-making, problem-solving and creativity. This collaborative AI, embodied as virtual or physical agents, fosters the development of mutualistic, symbiotic human-AI relationships, promoting partnership behaviors and collaborative cognitions. This MSc-course centers on theories, models and methods for developing such behaviors and cognitions, including value alignment, goal sharing, workload harmonization, co-learning and collaborative memories. Examples will be provided from diverse application domains, including childrens diabetes management, people with dementias activities of daily living (ADL) and first responders disaster management. Next to the lectures, students work in small groups on a specific Hybrid Intelligence design challenge, focusing on social robots that support people with dementia (see http://rejam.tudelft.nl). This way, you will become acquainted with the application of a coherent set of Socio-Cognitive Engineering (SCE) methods: - Value Alignment; Scenario-based Design; Claims Analysis; Design Patterns. - Social, Cognitive and Affective Modeling. - Prototyping; Evaluation; Research Ethics.	
Study Goals	At the end of this course, students will be able to: 1. Explain the core models and methods of Socio-Cognitive Engineering (SCE), including their strengths, weaknesses and complementarity. 2. Apply these models and methods in research and development projects, constructing a theoretically and empirically grounded design rationale and prototype. 3. Systematically build on state-of-the-art developments in an iterative process, explicating and testing the strengths and weaknesses of the design choices. 4. Explain which new insights are acquired by SCE for a specific Hybrid Intelligence (HI) development. 5. Write a structured report about a design-test cycle, with sufficient detail for a new group of researchers to continue the research. 6. Collaborate in a group to develop a HI-system, presenting intermediate and final results.	
Education Method	LECTURES During the lectures, the teachers will present a range of theories, models, and methods relevant to Socio-Cognitive Engineering (SCE). Students are required to read a number of scientific papers which are made available on Brightspace, along with the sheets/slides of the lectures. Together, the sheets/slides and the papers provide the students with the required theoretical knowledge to work on the practical project, and to learn about relevant design and evaluation methods, human factors theories, conceptual solutions, and design principles. Most of the lectures include practical assignments and discussions stimulating the students to apply the contents of the lecture to their project (also see Project). PROJECT In the project, students work in groups to apply the knowledge acquired during the lectures. Students are required to plan, execute, present, and report on a complete design cycle (i.e. design, prototype, and evaluation) for a given design problem. The SCE-assignment centers on a social robot for older adults with dementia, and their social environment (https://rejam.tudelft.nl). The objective of the social robot is to improve humans physical, social, cognitive, and emotional well-being. The students will use the Wiki Socio-Cognitive Engineering (WiSCE) tool to specify the design rationale and its evaluation, step-by-step (see also https://xwiki.ewi.tudelft.nl/). Throughout the course, students will give presentations about their progress, on the design and evaluation of their prototype. The instructors provide feedback on the WiSCE content and the presentations.	
Literature and Study Materials	Papers from scientific journals on Brightspace. Lecture notes on Brightspace.	
Assessment	The module assessment concerns the processing and application of the theory and methods; the construction of the design (rationale) and the evaluation; and the provision of the resulting concise and coherent report (including the lessons learned): The final grade of the course consists of the following components: Presentations (weighting 15%) Prototype (weighting 15%) Project report according to the prescribed format (weighting 60%) Individual reflection (weighting 10%) Final grade calculation: 0.15 * Presentations + 0.15 * Prototype + 0.6 * Project report + 0.1 * Individual reflection A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentations: resit Prototype: re-submit deliverable Project report according to the prescribed format: re-submit deliverable Individual reflection: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	There is no exam. The assessment is based on a paper, presentation and report. During the course, students will receive feedback on interim work. There is no resit after the end of the course.	

CS4295	Release Engineering for Machine Learning Applications	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	While the course does not have formal pre-requisites, some skills are expected: - You are able to use basic Git commands - You have a basic understanding of networking concepts - You are not afraid of using a command line interface The course will provide introductory material for self-study that can serve as a refresher or provide pointers for students to fill knowledge gaps.	
Course Contents	The world of Software Engineering has been revolutionized in the last decade. Instead of releasing software updates yearly, companies can now release multiple times per week, sometimes even per day, to their customers. This allows quick reactions to market demands, software failures, and it is crucial for increasing the business value of software. These improvements have been mostly enabled by advances in release engineering and, in this course, we will learn about the techniques and technologies that build the foundation for modern release engineering. This course is addressing two main target audiences: 1) software engineers that would like to learn how to systematically release their applications and make them scalable in an orchestrated environment and 2) machine learning experts that would like to learn how to make their proof-of-concept applications production ready. Together, we will go on a journey that starts at continuous integration and then moves on to continuous delivery, continuous deployment, and continuous experimentation. We will discuss the theory and the current research on various related subjects like containerization, testing, or monitoring and will put the learned theory into practice. As a running example, we will build a pipeline for a machine learning application, which -compared to traditional release engineering- poses additional challenges, like data versioning or model deployment. *Important:* Make sure to register for a group in Brightspace AT THE LATEST during the first lecture. *Important:* The course introduces several state-of-the-art tools that are used in industry. While -in theory- all major operating systems are supported, experience has shown that Windows users will face challenges. To reduce the friction points, we recommend affected students to either dual-boot a Linux system or to make use of a Linux Live system via USB when working on the exercises and assignments.	
Study Goals	After following this course, you will be able to... 1) Explain the concepts of DevOps and MLOps 2) Automate the development and operation process ... Automate packaging, versioning, and releasing in modern build environments like GitLab or GitHub ... Release artifacts in public registries like the GitHub Maven Registry. ... Provision servers with Vagrant and Ansible ... Automate the model training pipeline using DVC 3) Use modern containerization technology to build a scalable micro-service architecture ... Use Docker to isolate a complex, web-based application in a container ... Use Kubernetes to orchestrate a non-trivial, distributed application ... Create an Istio service mesh in Kubernetes that enables custom routing strategies ... Use Prometheus and Grafana to monitor the state of a Kubernetes-based application ... Find and understand documentation of open-source tools 4) Deploy and maintain machine learning models in production ... Embed a typical ML model in an HTTP micro-service ... Apply version-control techniques to data and models in a machine learning project ... Implement quality-control techniques in a machine learning application ... Design a testing strategy that covers data, model, infrastructure, and monitoring aspects 5) Document the state and extensions of release engineering pipeline ... Find related work and inspiration for release engineering tasks ... Identify extension opportunities in existing release pipelines	
Education Method	- Follow interactive lectures and pre-recorded material for self-study - Execute a programming project in small teams - Transfer lecture contents into a project through practical assignments - Assess the work of yourself and others through rubric-supported peer-feedback - Actively participate in Q&A sessions - Document your release pipeline in a report - Present your project results and answer questions about the lecture contents.	
Literature and Study Materials	- Lecture slides and recordings - In-class exercises - Tool documentation - External articles - Research papers	
Assessment	--- Formative Assessment --- You can check your individual learning progress through weekly rubrics ... You will receive feedback from other groups as a neutral outsider ... You can check your own progress via self-assessment Joint Q&A sessions will discuss common points of the peer feedback and concrete questions --- Individual Knock-Out Elements --- The course enforces several criteria to protect the active members of a team. Students not reaching the following criteria will be removed from the course:	

Upholding a professional attitude when interacting with peers and staff.
Actively participating in the team process and contributing to all team deadlines and deliverables
Constructive participation in all peer reviews for assignments (one for another team and one for your own)
Meaningful and visible project contribution throughout each of the seven project weeks (W2-W4, W6-W9)
... Create 1+ mergeable PR with meaningful* contribution
... Approve 1+ PR that is not your own
... Failing to satisfy the criteria in two subsequent weeks will lead to a course fail
At the end of the couse, you must meet the following criteria
... 7+ meaningful* PRs of you were merged to main
... You approved 7+ meaningful* PRs of others
... You did not miss peer reviews for more than one assignment.
Ability to explain lecture contents and defend all aspects of the team project in the oral examination. The inability to explain or justify the project in the oral examination can be repaired in a second oral examination.
Availability of team results for grading (repositories, report, artifacts)

(*) A "meaningful" contribution adds source code, configuration options, or report content. We do not count comments or whitespace.

The full specification of the knock-out criteria will be posted on Brightspace.

--- Summative Assessment ---

You can only get a grade for the course, if you pass the following pass/fail elements:

You have passed all knock-out criteria.
You passed the Presentation & Oral Examination

Your grade then consists of the following partial grades. ALL partial grades are rounded to .1 and must at least be "almost passed" (>=5.0). Their weighted average will be rounded to .5 to calculate the final team grade.

Versioning & Releases (10%)
Containers & Orchestration (15%)
Kubernetes & Monitoring (15%)
Provisioning (15%)
Istio Service Mesh (15%)
ML Configuration Management (15%)
ML Testing (15%)

In case of large performance differences in a group, we may choose to assess the team members individually, based on the individual contributions to the deliverables and the interaction traces that we find in the repository.

--- Please Note ---

All assignment deadlines throughout the course are fixed, we do not accept late submissions. Late submissions in systems that do not enforce due dates will be treated as not submitted (e.g., Brightspace, Buddycheck).
Failing to submit a deliverable cannot be repaired.
Insufficient deliverables (>=4.0 and <6.0) can be repaired, but the grade of the deliverable is then capped at 6.0.
There is NO resit opportunity for this course.
Partial grades are not carried over to the next academic year.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5)

EE4675	Object classification with radar	4
Responsible Instructor	Dr. F. Fioranelli	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Advised: ET4169, specifically knowledge of microwave and radar basics such as measurements of range and velocity and related basic signal processing.	
Course Contents	While the idea of gaining information about the surrounding environment using electromagnetic waves is a relatively mature concept (a radar essentially!), a tremendous amount of research activities have recently coupled the analysis of data from EM sensors with techniques that can teach them to recognise automatically objects of interest. What natural or manmade objects are moving around, and what are they doing? Can we infer something about their behaviour or intention from such movements? These are an example of the big questions which data and clever signal processing from EM sensors can address. Applications range from the analysis of human gait and movements to the discrimination of birds vs Unmanned Aerial Vehicles (drones), to the identification of the subtle gestures people produce to interact with smart objects, amongst others. All these applications have in common two main areas of expertise: one has to do with understanding and processing the information encoded into the sensors signals (radar signal processing and data format); one has to do with the techniques to interpret this information automatically to make decisions (we could call this applied machine learning). In this course, we aim to provide a comprehensive introduction and explanation of both areas from an engineering application perspective, with the objective to learn how to use these tools for real-life applications in radar engineering. The course is broadly divided into two parts: 1. Part 1 will build on fundamentals of radar & microwave sensing courses to investigate the concept of micro-Doppler signatures, focusing on its modelling, the signal processing for its characterisation, and its relationships with the physical movement of objects and therefore their identity. Can we understand from the EM sensors data how the different parts of human bodies or manmade objects move? 2. Part 2 will introduce machine learning techniques for extraction of information from sensors data (feature extraction), for their selection, and for classification of objects of interest. Conventional supervised learning techniques will be presented (for example Nearest Neighbours and Support Vector Machine classifiers), along with some of the more recent data-driven approaches based on neural networks. The diversity of formats of the radar data for classification will also be considered, including micro-Doppler signatures, High Resolution Range Profiles, radar cubes. In both parts, students will be exposed to concrete examples and data directly taken from current activities at TU Delft, so that the research element is well embedded in the teaching. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Study Goals	The aim of this course is to introduce students to techniques to classify and identify objects of interest mixing machine learning and signal processing applied to data from radar sensors. By the end of this course, students will be able to: -- Apply relevant signal processing techniques to EM sensors and radar data to extract micro-Doppler signatures -- Compare the different domains of radar data that can be exploited for objects classification, and the convenience of using one rather than the other in different situations -- Analyse machine learning techniques for feature extraction and classification applied to radar sensing -- Propose possible approaches for given classification problems based on radar sensors data	
Education Method	The course consists of class lectures and practical activities in the laboratory where possible or following the lectures. Laboratory activities may take place at the existing facilities available for the MS3 (Microwave Sensing Signals & Systems) section. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Course Relations	This course is intended as part of the Master program in Electrical Engineering, primarily for students in the WICOS and S&S tracks, but with expected interest also from students from the civil engineering faculty (CITG), robotics/mechanical engineering (ME), and aerospace (LR), where activities in remote sensing and autonomous robotics have seen recent significant interest in radar sensing and radar-based classification. This course is envisaged to cover new topics, currently not present in the courses related to radar sensing on offer. In this regard the course is complementary to the existing courses of ET4169 (Radar I) providing the fundamentals of radar sensing and radar as an instrument, and ET4175 (Radar II) providing advanced information mostly focused on the theoretical basis of processing for radar detection and tracking. Therefore, this proposed new course fills a gap in presenting advanced content on the topic of radar-based classification and the related approaches, widening the offer of topics available to our students and contributing to their more well-rounded education. This new course can fit well with the new proposed order of the radar courses": EE4565 in Q2 (fundamentals of electromagnetic scattering and propagation, ET4169 in Q3 (basics of radar for broad audience of students), this course in Q4 (more specialised course in classification based on radar), ET4175 (more specialised course in advanced theory for radar detection and tracking). The new course also fits with the wider research strategy of the department, whereby classification and machine learning intersect the three overarching research themes of Next generation sensing and communication, Health and wellbeing, and Autonomous sensor systems. It is therefore expected that this course will enhance the research capabilities of students in view of their thesis projects, and benefit their employability afterwards. Specifically, radar-based classification for situational awareness is a timely topic, relevant to automotive (NXP), defence/security (Thales NL, TNO), robotics and HCI (Google has recently introduced a radar, Soli, within their phones for gesture-based interaction), healthcare (contactless estimation of vital signs). As part of the educational activities in this course, it is envisaged to have at least one guest lecture on relevant topic from an external stakeholder (for example from Thales NL and TNO).	
Literature and Study Materials	-Lecture slides and pre-class activities provided by the instructors -Selected chapters of the following books. They should all be available as e-book via the TU Delft Library. ---Richards, Mark A. Scheer, James A. Holm, William A.. (2010). Principles of Modern Radar, Volume I - Basic Principles. SciTech Publishing. ---Chen, Victor. Micro-Doppler Effect in Radar, Artech House, 2011. ---Chen, Victor C.; Tahmoush, David; Miceli, William J.. (2014). Radar Micro-Doppler Signatures: Processing and Applications. Institution of Engineering and Technology. ---Chen, Victor C.; Ling, Hao. (2002). Time-Frequency Transforms for Radar Imaging and Signal Analysis. Artech House. ---Gurbuz, Sevgi Zubeyde. (2021). Deep Neural Network Design for Radar Applications. Institution of Engineering and	

	Technology. ---Fioranelli, Francesco; Griffiths, Hugh; Ritchie, Matthew; Balleri, Alessio. (2020). Micro-Doppler Radar and its Applications. Institution of Engineering and Technology. ---Geron Aurelien. (2019). Hands-on machine learning with scikit-learn, keras, and tensorflow : concepts, tools, and techniques to build intelligent systems (Second). O'Reilly Media. -Selected published research articles available in IEEE Xplore (to be updated year by year, as novel research is generated - these are accessible via TU Delft Library)
Assessment	The final grade of the course consists of the following components: -a written exam with questions (that can be performed online in Brightspace if needed). -a report or oral presentation on practical/project activities (which may include collection and/or processing of some radar data, and implementation of a complete radar classification pipeline). The written exam is worth 75% of the final grade and the report(s) on practical activities 25%. As repair possibility the written exam has one resit opportunity. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).

ET4117	Electrical Machines and Drives	4
Responsible Instructor	Dr. G.R. Chandra Mouli	
Responsible Instructor	Dr.ir. H. Vahedi	
Instructor	W. Shi	
Responsible for assignments	H.J.M. Olsthoorn	
Responsible for assignments	H. Dialani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2014 (7th edition) ISBN 9781259254666	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovyi	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ME41006	Musculoskeletal Modeling and Simulation	4
Responsible Instructor	A. Seth	
Instructor	Ir. J.D.C. Groenhuis	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Human and animal movement is often smooth, powerful, and graceful. Animal movement is generated through a cascade of nervous, muscular, and skeletal system dynamics. Even the simple tasks of walking and running require the coordination of hundreds of muscles that pull on bones that move our limbs and generate forces on the environment so we can move. When these systems are disrupted through disease and injury, they result in movement disorders that can be debilitating. In this course you will learn to construct and simulate musculoskeletal models that generate movement. You will learn how to apply models to improve measurements of movement and to predict behavior, thus extracting more insights about human (and animal) movement than from observations alone. We will start with the constituent components of bones, joints, muscles, and tendons and their dynamics, as the building blocks of musculoskeletal systems. You will learn inverse techniques to extract unmeasured quantities, like joint kinematics, work, and power from observations and to use forward dynamic simulations to predict behaviors and answer what if? type questions. You will apply these models and techniques to answer a research question of your own formulation	
Study Goals	The goal of this course is to teach you how to use state-of-the-art computational tools to model and simulate the human musculoskeletal system. Learning will be guided by lectures, reading articles, practical labs and a final project. Upon successfully completing this course, you will be able to: 1. Model Develop and analyze mathematical models of the human musculoskeletal system using OpenSim and MATLAB. 2. Simulate Create inverse and forward dynamic simulations of human movement. Perform inverse dynamic simulations and optimizations using experimental data to compute quantities that cannot be easily measured (e.g., muscle forces). Perform forward dynamic simulations to reproduce observed movements and predict new movements. 3. Analyze Use musculoskeletal simulation to answer questions about human movement (e.g., how does a muscular impairment affect muscle coordination?), and validate results against experimental data. 4. Critique Read journal papers in the field; establish the main contributions and limitations of the work. Summarize your work using a thoughtful structure and relate your work to the broader field. You will also review the work of your peers and provide feedback on clarity, correctness and impact. 5. Formulate a research question and write a corresponding proposal.	
Education Method	Flipped Classroom method. Lecture slides, reading materials, and assignment descriptions provided through Brightspace. Class time is used to review reading materials and answer questions on concepts and practical labs. The bulk of the work is performing challenge based labs and a final project of your choice.	
Assessment	60% Three practical computer lab assignments (20% each) 40% Final project (proposal, presentation, abstract, reproducibility, reviews) Pass 6/10 grade total	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

ME41056	Multibody Dynamics	5
Responsible Instructor	J.K. Moore	
Instructor	W. Wolfslag	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	Vector calculus, Physics, Linear algebra (e.g. Wiskunde 2 Lineaire Algebra WBMT1051), Differential equations (e.g. Wiskunde 3 Differentiaal vergelijkingen WBMT2048 T2), Numerical methods (e.g. Wiskunde 4 Numerieke Wiskunde WBMT2049 T2), Statics (e.g. Statica WB1630-20), Dynamics (e.g. Introductory Dynamics WB1135 and/or Rigid Body Dynamics WB2630), Scientific computing (Python, Matlab, Julia, etc.)	
Course Contents	Multibody systems are mechanical systems made up of multiple interconnected rigid bodies. In this course, you will learn a systematic approach for generating, solving, and integrating equations of motion used to model multibody systems. Topics covered are: - Three dimensional kinematics and use of Euler angles and Tait-Bryan angles for describing orientation. - Equations of motion for planar and three-dimensional systems, free body diagrams, constraint equations and constraint forces, uniqueness of the solution. - Systematic approaches for generating equations of motion, e.g. Newton-Euler, Lagrange, coordinate transforms (TMT), Kane, etc. - Relationship of maximal coordinate equations of motion to generalized coordinate equations of motion. - Configuration (holonomic) constraints, e.g. closed kinematic loops, and motion (nonholonomic) constraints, e.g. rolling without slipping, and their relationship to degrees of freedom. - Symbolic and numeric computational methods applicable to generating and solving equations of motion. - Numerical integration of coupled differential and differential algebraic equations with careful attention to stability, accuracy, and minimization of integration drift. - Visualization and analysis of two- and three-dimensional motion.	
Study Goals	The student is able to find the motions of linked rigid body systems in two and three dimensions including systems with various kinematic constraints such as sliding, hinges, rolling, and closed kinematic loops. More specifically, the student must able to: 1. derive the equations of motion for a simple planar system, draw free body diagrams, set up constraint equations, and demonstrate the uniqueness of the solution 2. derive the equations of motion for a system of interconnected rigid bodies by means of a systematic approach 3. derive the equations of motion in terms of generalized independent coordinates 4. apply the techniques from above to systems having holonomic and nonholonomic constraints 5. perform various numerical integration schemes on the equations of motion, and predict the stability and accuracy of the applied methods 6. perform numerical integration on a coupled system of differential and algebraic equations (DAE's) with minimal constraint drift 7. derive the equations of motion for a general rigid three-dimensional body system bound by constraints 8. identify the need to describe orientation in 3D space by means of: Euler angles and Tait-Bryan angles and derive the angular velocity and accelerations in terms of these parameters and their time derivatives 9. create accurate, efficient, and documented computer programs that construct and simulate multibody systems 10. analyze and interpret the resulting motion from a multibody simulation	
Education Method	Lectures, homework, and work sessions	
Computer Use	Laptop needed in work sessions	
Assessment	Weekly assignments and an exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

ME41065	System Identification and Parameter Estimation	7
Responsible Instructor	M.L. van de Ruit	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic control engineering,s signal analysis and experience with Matlab	
Course Contents	System identification deals with the estimation of physical properties of (dynamic) systems from measured data (inputs and outputs), which is essential for the understanding of system behavior. System identification is particularly useful for biological systems where physical entities are generally not known a priori. The course provides mathematical a background of system identification in both time domain and frequency domain with a strong focus on practical application. Major topics to be studied are: spectral densities, nonparametric black-box models, discrete time domain models (parametric grey-box), closed loop systems (utilizing feedback), translation of transfer functions into physical parameters, optimization routines and design of perturbation signals. During the course examples from both technical and physiological systems (i.e. estimate the behavior of a driver or the dynamics of a human joint) will be discussed.	
Study Goals	Part 1: Signals: At the end of the course you can. LO1a. explain the importance of signal acquisition properties for the identification of system dynamics. LO1b. explain the importance of input signal design for the identification of system dynamics. LO1c. apply your knowledge of input signal design and signal acquisition properties for successful system identification using MATLAB and Simulink. Part 2: System identification At the end of the course you can. LO2a. illustrate how system dynamics can be accurately estimated using non-parametric (non)linear time-invariant models in the time- and frequency domain. LO2b. apply your knowledge to estimate nonparametric models of unknown systems from measured input and output signals using MATLAB and Simulink. Part 3: Parameter estimation: At the end of the course you can. LO3a. explain what model structure best fits a non-parametric estimate. LO3b. explain which optimization algorithm to use to find the model parameters. LO3c. apply your knowledge to parameterize physical models from nonparametric estimates employing optimization techniques using MATLAB and Simulink.	
Education Method	Self study, lectures (2x2 hours per week) including consultancy for (homework) assignments	
Prerequisites	recommended prior knowledge - Probability theory and statistics (e.g. WBMT2049-T1) - Signal analysis (e.g. WB2235) - Systems and control engineering (e.g. WB3240)	
Assessment	Individual: Final closed book exam (60%), 3x digital test (total: 40%) Small groups: assignments (pass/fail) To pass the course, the final grade must be a 5.75 or higher.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Analysis Mathematics Matlab Mechanics Modelling Numeric Methods Programming Signals Signals and Systems Small groups Technology	
Department	ME Department Biomechanical Engineering	
Contact	By email SIPE-ME@tudelft.nl	

ME41085	Biomechatronics	4
Responsible Instructor	Dr.ir. G. Smit	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Biomechatronics is a contraction of biomechanics and mechatronics. In this course the central focus is on function and coordination of the human motion apparatus, and on the design of assistive devices for the support of the function of the motion apparatus. Examples are assistive devices like an orthosis, a prosthesis or Functional Electrical Stimulation of muscles. The goal is to provide some function to patients with functional deficiencies.	
Study Goals	After successfully completing this course, you are be able to: 1. make a Problem Analysis of a given assignment in the field of the human motion apparatus and its interaction with an assistive device a. identify the underlying cause [pathology] of the problem as presented in the assignment b. describe and explain the possible treatment options for the pathology of 1a c. translate the result of 1a into technical Design Objectives without reference to any solutions d. derive qualitative and quantitative Design Specifications from the Design Objectives, and categorize and prioritize these 2. optimize the assistive device application given in the assignment in energetic and control aspects a. select and apply appropriate Design Methodology and Design Methods b. generate a variety [typically at least three] of Conceptual Designs c. judiciously select the most appropriate Conceptual Design d. demonstrate the plausibility or feasibility of the Conceptual Design, with special emphasis to the patient benefits	
Education Method	Lectures (2 hours per week)	
Literature and Study Materials	Course material: References from literature: D.H. Plettenburg Upper Extremity Prosthetics. Current status & evaluation VSSD (2006) ISBN-13: 978-9071301759	
Prerequisites	Assumed prior knowledge when entering the course, a basic understanding of: Statics Dynamics Control Theory Engineering Design	
Assessment	Assignment: Report made in a group (~2 persons)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: WB2432	
Department	ME Department Biomechanical Engineering	

ME41096	Bio Inspired Design	5
Responsible Instructor	Prof.dr.ir. P. Breedveld	
Instructor	B. van Vliet	
Contact Hours / Week x/x/x/x	4.4.0.0	
Education Period	1 2	
Start Education	1	
Course Language	English	
Expected prior knowledge	Mechanics and basic design experience in hand sketching and CAD software, like SolidWorks.	
Course Contents	The course Bio-Inspired Design gives an overview of non-conventional mechanical approaches in nature and shows how this knowledge can lead to more creativity in mechanical design and to better (simpler, smaller, more robust) solutions than with conventional technology. The course discusses a large number of biological organisms with smart constructions and unusual internal mechanisms and presents a number of technical examples and designs of bio-inspired (surgical) instruments, devices, robots, vehicles and machines. Examples of topics: Life and its creation, engineering in biology, strength at low weight, stiffness with soft structures, robustness and redundancy, storing energy in compliant springs, energetically efficient muscle configurations, clamping with hands, claws, suction, glue, dry - and wet adhesion, biological vibration systems, biological walking, swimming, crawling and flying, locomotion of micro- and single-celled-organisms, biological sensors, soft robots, walking robots, flying robots, snake-like surgical instruments and devices. To teach students how to create smart and truly innovative designs, students are trained with the ACCREx creative design method that was developed at the ME department Biomechanical Engineering. Alternating intuitive brainstorming with logical, scientific abstracting and categorizing, ACRREx structures and guides a design process towards fundamentally new design solutions. Structure of the course: 1. Engineering & biology - Life and its creation - Finding biological information - Creating creativity 2. Bioconstruction (how are creatures constructed) - Mechanical stiffness & motion - Hydrostatic stiffness & motion 3. Biopropulsion (how do creatures move) - Flying at macro- & microscales - Floating, rolling & adhesive devices - Sliding, crawling & snake-like devices - Walking, compliant devices & compliant actuation 3. Bioclamping (how do creatures grasp) - Friction & biological grasping - Biological hands & prosthetic hands - Biological adhesion	
Study Goals	The student must be able to: 1. describe methods for creative design 2. identify mechanical working principles and phenomena of biological creatures - explain their construction, motion, and/or processing mechanisms - formalize the essence of these mechanisms in models - derive non-conventional design principles from these models 3. implement these design principles in innovative mechanical devices - summarize the transition process from the biological to the mechanical domain - present their design in drawings or preferably in working models	
Education Method	Interactive lectures Interludia with driven scientists in a talkshow-like setting Student presentations about the design process Reflective Q&A sessions	
Assessment	Students are subdivided into multidisciplinary student groups. Each student group receives a design assignment for which a novel, bio-inspired solution has to be found. During the design process, the student groups are coached by the teachers and have to give a number of presentations about their progress to the other student groups. The examination consists of three parts: 1. Each student group hands in a scientific paper covering a problem analysis, the biological background, the design process, the final solution and an evaluation. 2. Each student group gives a final presentation to the teachers and the other student groups. 3. Each student group fabricates a demonstration model that shows the essence of the final solution. The final mark of each individual student is based on a weighted average between these group results and his/her functioning within the student group.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

ME41125	Introduction to Engineering Research	3
Responsible Instructor	Dr.ing. L. Marchal Crespo	
Instructor	Drs. A.E. Kam	
Instructor	L. Hoogendam	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	**This course will not be taught this year** Students must have have basic programming skills in order to perform simulations or process data, for example in Matlab, Python, or R. The ability to derive the equations of motion and to simulate and design a controller for a mechanical system is helpful for this course.	
Course Contents	1) Literature review 2) Academic Writing 3) Dealing with uncertainty in experimental data 4) Generating reliable results 5) Preparing data and code for FAIR use 6) Reviewing 7) Dealing with feedback 8) Oral presentation	
Study Goals	Learning objectives By the end of this course you should be able to: 1) review the literature on a chosen topic 2) write a formal academic paper following the standard structure elements of an academic paper about an engineering topic 3) analyze the validity of data and statistical analyses 4) generate reliable simulations and code 5) assemble your data and your code for FAIR use 6) assess peers work using criteria provided by the instructor 7) revise an academic text, using the lecturers and/or peer feedback 8) present your own research to a mixed audience	
Education Method	Lectures, active learning activities, material provided on Brightspace, assignments.	
Assessment	Assessment is based on assignments and the final presentation, as follows: 1) 25%: your first draft academic paper 2) 25%: quality of the feedback provided to others (in two rounds) 3) 25%: rebuttal letter and final version of your paper. 4) 25%: your final presentation.	
Enrolment / Application	The course will only be offered if a minimum of 20 participants is reached. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

ME44206	Quantitative Methods for Logistics	5

Responsible Instructor	Dr. B. Atasoy
Instructor	Ir. M.B. Duinkerken
Contact Hours / Week x/x/x/x	2/2/0/0
Education Period	1 2
Start Education	1
Exam Period	2 3
Course Language	English
Course Contents	The course contains of two parts; roughly 7 weeks each. In the first part of the course, an introduction to basic operations research techniques will be presented: - linear programming (simplex, duality etc.) - integer programming (IP, BIP, MILP etc.) - basic probability and statistics; brief overview of models to deal with uncertainty This part of the course will largely be based on the first chapters of the book by Hillier and Liebermann. During the first quarter, students must complete an individual practical assignment where they develop a mathematical model and implement it using Python together with Gurobi as a solver. In the second part of the course, several common quantitative methods for transportation and logistics are presented. - networks and routing: network flow problems, facility location problems, transportation problem, transshipment problem, vehicle routing problems and other standard problems; standard algorithms i.e. the branch and bound method. - queueing theory: an overview of basic models and results. During the second quarter, students must complete a practical assignment in groups on a case study with a transportation network again using Python.
Study Goals	At the end of this course, the students will be able to: - Explain the importance and use of operations research techniques to analyze and improve systems (i.e., linear programming, integer programming, routing, scheduling, queueing) - Formulate mathematical models of real-life problems (e.g., resource allocation, production planning, facility location, transportation) with operations research techniques - Perform verification and validation on mathematical models - Evaluate operation research techniques based on strengths and weaknesses (in terms of accuracy and efficiency) for improving real-life problems - Apply basic operation research techniques to small-scale problems as well as realistic case studies
Education Method	Flipped classroom approach: The students are expected to be prepared by studying the online material that is indicated every week. Lectures in two parts: (1) First 45 min course content, (2) Second 45 min assignment - Course content includes a short intro and exercises related to the subject. - Assignment part is dedicated to the introduction of the software (Python+Gurobi), the introduction of the assignments, and any further needed assistance and feedback. It is strongly recommended to bring a laptop to the lectures. There will also be additional reading material that will be provided during the course.
Books	Hillier & Lieberman. Introduction to Operations Research. 10th Edition, McGraw-Hill, ISBN 9781259253188 Additional resource (also online): From theORy to application: learning to optimize with Operations Research in an interactive way https://doi.org/10.59490/tb.94
Assessment	Practical Q1 - 25% (individual: Mathematical Model, Python implementation and report) Practical Q2 - 25% (group: Mathematical Model, Python implementation and report) Exam Q2 - 50%. For all 3 parts a minimal grade of 5 is required. The exam will be given at the end of Q2, the resit for the exam in Q3 Both assignments are only given once a year; with a limited option to resubmit if the grade is insufficient. The highest grade for a resubmission is 6. In case of measures resulting from COVID-19, the prescribed assessment will be an online examination. If the opportunity arises for examinations on campus, the method of assessment will be on campus.
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Remarks	Be aware of a significant overlap regarding the content between CIEQ6112 (Transport Engineering & Optimization) and ME44206 (fundamental for TIL and MME). Therefore, for ME-MME students and TIL students full credit might not be given to CIEQ6112 as an elective. Contact the program coordinator for details.
Department	ME Department Maritime & Transport Technology

ME44312	Machine Learning for Transport and Multi-Machine Systems	3
Responsible Instructor	F. Schulte	
Instructor	Dr. B. Atasoy	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic programming	
Course Contents	The course gives an introduction to the basic concepts of machine learning (ML) (supervised, unsupervised, reinforcement learning) and puts them into relation to artificial intelligence and data analytics. Important fundamental machine learning algorithms including regression, classification, clustering, and neural networks are introduced and implemented in Python. The course also gives a brief overview of more advanced ML techniques that are used in the transportation and multi-machine domains such as reinforcement learning, graph-neural networks, and recommender systems. On this basis, machine learning methods are applied in group projects to transport and multi-machine systems using different types of data from different domains (e.g., AIS, GPS, New York Taxi, or machine failure data).	
Study Goals	-Understand the basic concepts of machine learning -Learn to implement widely-used machine learning algorithms -Apply machine learning methods to transport and multi-machine systems -Evaluate different machine learning methods for a problem setting -Conduct small ML projects	
Education Method	-Lectures -Group Work -Individual exercises for each machine learning algorithm	
Computer Use	Python on laptop	
Assessment	Group report	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Abstract Algorithmics Artificial intelligence Broad Business Design Group work Industry Modelling Optimalisation Programming Research Methods Stochastics Sustainability Technology Transport & Logistics Transport phenomena	
Percentage of Design	40%	
Design Content	Designing the right machine learning methodologies for different problem settings in the context of transport and multi-machine systems	
Department	ME Department Maritime & Transport Technology	
Contact	f.schulte@tudelft.nl b.atasoy@tudelft.nl	

ME46055	Engineering Dynamics	4
Responsible Instructor	Dr. F. Alijani	
Instructor	R.A. Norte	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Multibody Dynamics A(ME41050), Multibody Dynamics B (ME41050), Nonlinear mechanics(ME46000), Nonlinear dynamics (ME46072).	
Course Contents	The dynamic behavior of structures (and systems in general) plays an essential role in engineering mechanics and in particular in the design of controllers. In this master course, we will discuss how the equations describing the dynamical behavior of a structure and of a mechatronical system can be set up. Fundamental concepts in dynamics such as equilibrium, stability, linearization and vibration modes are discussed. If time permits, also an introduction to numerical integration techniques is proposed. The course will discuss the following topics: --Review of the virtual work principle and Lagrange equations --Linearization around an equilibrium position: vibrations --Free vibration modes and modal superposition --Forced harmonic response of nondamped and damped structures -- Introduction to time integration techniques	
Study Goals	The student is able to select different ways of setting up the dynamic equations of mechanical systems, to perform an analysis of the system in terms of linear stability and vibration modes and to properly use mode superposition techniques for computing transient and harmonic responses. More specifically, the student must be able to : 1. Explain the dynamic principle of virtual work and Lagrange equations, and discuss their relation to the basic Newton laws. 2. Describe the concept of kinematic constraints and identify a proper set of degrees of freedom to describe a dynamic system. 3. Use Lagrange equations via employing the minimum set of degrees of freedom to find the governing equations of dynamic systems, and construct these equations for systems with kinematic constraints. 4. Use Hamiltons principle to find the governing equations of motion of dynamics systems. 5. Find equilibrium positions and construct the linearized equations of motion by using different contributions of the kinetic and potential energies. 6. Analyze the linear stability of dynamic systems according to their state space formulation. 7. Explain and use the concept of free vibration modes and frequencies for multi degree of freedom systems. 8. Apply the orthogonality properties of modes to describe the forced response of damped and undamped systems.	
Education Method	Lecture	
Literature and Study Materials	Course material: Lecture notes (available through brightspace) References from literature: 1. M. Géradin and D. Rixen. Mechanical Vibrations. Theory and Application to Structural Dynamics. Wiley & Sons, 2nd edition, 1997. 2. J. Ginsberg. Engineering Dynamics. Cambridge University Press, 2008. 3. D.J. Inman, Engineering Vibration. PrenticeHall, 1996. 4. L. Meirovitch. Principles and Techniques of Vibrations. PrenticeHall, 1997.	
Assessment	written exam +A quiz or an assignment	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	In case an assignment is given- The assignment will involve the use of a Matlab-like software 2. Old course code: WB1418-07	
Department	ME Department Precision & Microsystems Engineering	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Prof.dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Dr. O. Nejadseyfi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be 5.8 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest decimal. The maximum bonus score is 0.8. The bonus only applies to reports with grade >= 5.5 and submitted before the first report deadline.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	ME Department Precision & Microsystems Engineering	
Contact	Prof.dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory Fundamentals of Coding in Python and MatLab	
Course Contents	Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Symbolic regression * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control We will aim for a balance of practical intuition and theoretical understanding: we will discuss examples of applications and go deeper into the theory when feasible.	
Study Goals	 At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures (recorded and in class) and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	 Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: (i) practical assignment and (ii) written exam, open book. You must score at least 5.5 on both assignments to get a pass. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity to resit if they fail the written examination. * If your practical assignment is graded between 5 and 5.75, you will have two weeks to integrate the assessors feedback. If this is done satisfactorily, you can be rewarded with a mark that is not higher than 6/10. The exact instructions for the addition will be announced after the grading of the PA is concluded.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding course registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) exam registration is obligatory. More information regarding exam registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Artificial intelligence Complex Mathematics Intensive Lineair Algebra Mathematics Modelling Programming Project Proofs	
Department	ME Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42075	Modeling and Control of Hybrid Systems	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	core Systems & Control topics (including state-space models, discrete-time and continuous-time models, stability, linearization), optimization basics (linear programming), Matlab	
Course Contents	Recent technological innovations have caused a considerable interest in the study of dynamical processes of a mixed continuous and discrete nature. Such processes are called hybrid systems and are characterized by the interaction of time-continuous models (governed by differential or difference equations) on the one hand, and logic rules and discrete-event systems (described by, e.g., automata, finite state machines, etc.) on the other. A hybrid system also arises in practice when continuous physical processes are controlled via embedded software that intrinsically has a finite number of states only (e.g., on/off control). This course provides an introduction to modeling, analysis, and control of hybrid systems. The main topics considered are: * General introduction, examples of hybrid systems and motivation * Modeling frameworks (automata, hybrid automata, piecewise-affine systems, complementarity systems, mixed logic dynamical systems, Petri nets) * Properties and analysis of hybrid systems (well-posedness, Zeno behavior, stability, liveness, safety, ...) * Control of hybrid systems (switching controllers, model predictive control) * Verification and tools	
Study Goals	After successfully completing the course, you will be familiar with the main aspects and characteristics of hybrid systems, you have insight in trade-off modeling power versus decision power in the context of analysis and control of hybrid systems, and you will have obtained an overview of modeling, analysis, and control methods of tractable classes of hybrid systems. In addition, you will be able to apply hybrid systems modeling and control to simple case studies.	
Education Method	lectures + mandatory assignment Attending the lectures is not mandatory, provided the recorded lectures and lecture materials are processed via self-study.	
Assessment	Written exam (individual, closed-book, no calculators, counts for 60% of the final marks) + assignment (group assignment, assessed through written report, counts for 40% of the final marks) To successfully pass the course the final weighted average score for the exam and the assignments together should be at least 5.75. Important: partial marks for exam or assignment do not carry over from one academic year to the next.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	
Contact	Questions are preferably asked during, before, or after the lectures, or via the Discussion forum on Brightspace.	

SC42101	Networked and Distributed Control Systems	4
Responsible Instructor	Prof.dr.ir. T. Keviczky	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course starts with the modelling of so-called networked control systems (NCS). Networked control systems are systems in which the communication between plant and controller takes place via a (e.g. wireless) network. Such network-based communication leads to imperfections in sensor and control signals, such as time-varying and uncertain sampling intervals and delays, packet dropouts, scheduling constraints, etc. The modelling framework introduced in the course is subsequently used to support stability analysis, using characterizations based on linear matrix inequalities (LMI). Such stability analysis allows to study trade-offs between requirements on the controller, the network and plant properties. The second part of the course deals with the aspect of the distributed control of networked systems. In particular, distributed optimization methods and various decomposition techniques (primal, dual, augmented Lagrangian / proximal point method, ADMM), links to consensus algorithms, and their application in networked multi-vehicle distributed robotics problems. Online optimization-based control approaches such as distributed model predictive control for multivehicle cooperation, distributed LQR and decomposition based methods that are applicable to collections of mobile agents. The methods will be illustrated on application examples including cooperative rendezvous, distributed formation control, spacecraft formation flight, and robotic networks.	
Study Goals	The student must be able to: 1. create models for networked control systems with network-induced uncertainties / effects / imperfections, including time-varying and uncertain sampling intervals, delays, packet dropouts, and scheduling constraints 2. analyse the stability of NCS (involving the above effects), e.g. by applying LMI-based stability characterizations 3. describe and apply decomposition techniques for distributed optimization problems 4. describe and apply consensus algorithms to multi-agent coordination problems 5. solve cooperative control problems by implementing a distributed model predictive control approach 6. analyse and evaluate the stability and convergence of distributed optimization and control methods 7. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback	
Education Method	Lectures.	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course. Python is also allowed for some exercises/assignments.	
Literature and Study Materials	Course material: Lecture slides including additional reading material in the form of papers and textbooks are made available online.	
Assessment	1. This course is graded by four written assignment reports, each representing 25% of the final grade. 2. In order to complete the course, each assignment must be completed with at least a mark of 5,0 and the corresponding assignment report submitted before the deadline. In the event of a late submission, or a lower than 5,0 mark for any of the reports, no final mark will be given.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42110	Dynamic Programming and Stochastic Control	5
Responsible Instructor	M.A. Sharifi Kolarijani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Solid knowledge of undergraduate probability (e.g., conditional distribution, expectation, variance), linear algebra (e.g., eigendecomposition and Jordan forms), and calculus (e.g., gradient and extrema of a function). Mathematical maturity and the ability to provide precise and rigorous arguments are also important. A class in analysis will be helpful, although this prerequisite will not be strictly enforced.	
Course Contents	The course covers the basic models and solution techniques for stochastic optimal control problems and sequential decision-making under uncertainty. The course starts with a class of stochastic processes known as Markov chains. We then extend the model to include control variables (Markov Decision Processes, or in short MDPs) and study optimal control of a dynamical system over a finite number of stages. We also discuss an important class of problems known as optimal stopping problems. Different applications from engineering and finance to operations research and management science will be covered.	
Study Goals	After successfully completing this course, you will be able to - model real-world control and decision-making problems under uncertainty and in dynamic environments using the framework of discrete-time Markov decision process; - compute relevant technical concepts including hitting probabilities, expected hitting times, invariant distributions, and the long-run proportion of time spent in a given state; - formalize the optimal decision-making problem by deriving the respective dynamic programming principle; - use the dynamic programming algorithm to find useful properties of the desired solution and solve the optimal decision-making problem.	
Education Method	Lectures and tutorials	
Literature and Study Materials	- The lecture notes of the course (available on Brightspace) - Dynamic Programming and Optimal Control, 3rd edition, D. Bertsekas, Athena Scientific, 2005 - Introduction to Probability, D. Bertsekas and J. Tsitsiklis, Athena Scientific, 2002	
Assessment	Written exam (If on-campus exams will not be possible, then all exams will be organized online via Ans)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	
Contact	The students can reach the course staff via email regarding administrative issues.	

SC42125	Model Predictive Control	4
Responsible Instructor	Dr.ing. S. Grammatico	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Model Predictive Control (MPC) is perhaps the most effective optimal control strategy for constrained dynamical systems. The basic concept of MPC is to exploit a dynamic model to forecast the system behavior, and optimize the forecast to determine the control input as the best decision at the current time. With emphasis on constrained linear systems, the course will present the theoretical fundamentals of MPC, such as Lyapunov stability, optimality and robustness, and its computational methods, such as quadratic programming.	
Study Goals	- Derive state prediction matrices from discrete-time linear models - Design the cost function, state and input constraints - Design MPC controllers with guaranteed recursive feasibility and asymptotic stability via appropriate terminal cost and constraint set - Formulate and solve constrained-linear-quadratic MPC problems via quadratic programming - Implement and simulate closed-loop systems controlled by MPC on Matlab - Design MPC controllers with integral action for reference tracking	
Education Method	Lectures, instructions.	
Assessment	Final project and oral exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Mathematics Matlab Optimalisation	
Department	ME Department Delft Center for Systems and Control	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	After this course you are able to: - recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - solve the shortest path problem and the transportation problem as well as some small flow shop problems. - solve the travelling salesman problem by the Branch and Bound algorithm. - apply several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - reproduce basic theorems concerning the mentioned problems and is able to prove some of these theorems. - explain how to solve large scale problems, especially shortest path and vehicle routing problems. - define scheduling problems and complexity theory. - implement the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam and a written resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Elective	Yes	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr.ir. J.T. van Essen	

WI4156(TU)	Game theory	6
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Open to all MSc students with a strong command of mathematics. Ideally suited for math students with an interest in finance.	
Expected prior knowledge	Standard BSc math courses: calculus, linear algebra, probability.	
Course Contents	This is a course in "mathematical" game theory, which is the counterpart of "algorithmic" game theory (agent based modelling, this part of game theory is taught in the computer science department). We cover all subjects from combinatorial games to cooperative games.	
Study Goals	Understanding the mathematical theory of rational behaviour/conflicts/negotiations/social choice. Game theory is one of the most prominent economic theories: three recent Nobel prizes have been awarded to game theorists. We will cover a very wide area, ranging from pure mathematics to economics, with an emphasis on the mathematical aspects. This is a course in the Finance Track of the Applied Maths programme, but it is open to all MSc students (math and non-math).	
Education Method	Lectures	
Literature and Study Materials	Game Theory - Thomas Ferguson - UCLA: find it on the net.	
Books	T.S. Ferguson, A course in game theory, World Scientific, 2020	
Assessment	The final grade of the course consists of the following components: written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students are expected to have: superior knowledge and skill in elementary probability (details below); experience working with Python, Matlab, or R; and a mathematical inclination. If this does not apply to you and you still want to enroll: contact the instructor. Good skills in elementary probability are indispensable as they are the foundation for the more advanced material of the course. We build on the concepts, tools, and methods of elementary probability, and students taking this course must be (or rapidly become) black-belt masters in these foundations, specifically: Chapters 2-5, 7, 9, and 10 of Dekking et al; prior knowledge of Chapters 13 and 14 is helpful, though we will revisit these subjects during the course. Testing yourself and doing a thorough review in the first few weeks is required; we will also spend some time on this in class to help you do this efficiently; after this a diagnostic test (carrying part of the course grade) will assess your mastership; sub par performance means you cannot follow the course.	
Course Contents	This course on Monte Carlo simulation and stochastic processes consists of a mixture of these components: modelling and models (and some of their pluses and minuses), learning a number of stochastic processes and some of their properties, doing Monte Carlo simulations, analyzing the generated data and drawing conclusions from them. Contents and format of this course are intended to combine the following: 1. real (possibly stylized) problems as starting points; 2. enough about modelling to make participants aware of the impact of modelling decisions; 3. identification of a number of stochastic processes present in/related to models and problems; 4. (limited) treatment of underlying theory, guided by: (a) what is necessary to ensure that participants know how to do things right and (b) what enables them to do things smartly; 5. facts, properties, and theorems: (a) sincere attempt to understand where possible; (b) focus on practical implications and uses; (c) illustrations rather than proofs or theory; 6. useful tools and techniques from statistics, probability, and stochastic processes; 7. simulation as aid to: obtaining answers, intuitive understanding, illustration of theory; 8. providing as much opportunity as feasible for participants to work on/with problems from their own field. Tentative topics/contents list: Review basic probability: most to be done as self-study, some in class, plus a mandatory test (see Assessment). Models, modelling. Simulation: Convert a model to a program and run it. How to expose and locate errors. Data, empirical distribution; estimation. Accuracy, efficiency. Sensitivity. Rare event simulation. Experimental design. Conditional distributions and expectations; conditioning as a tool. Multivariate distributions: general (some examples); multivariate normal. Stochastic processes: Random walk. Poisson process; renewal processes. Markov chains: discrete time; continuous time; long-term behaviour; embedded Markov chain; birth and death process; Brownian motion. A snippet of diffusions, (solutions to) stochastic differential equations (SDEs).	
Study Goals	Learning objectives: after following the course, participants will be able to 1. in an iid setting, carry out Monte Carlo simulations; and 2. extract and collect, summarize, and interpret the data thus generated; and 3. assess accuracy/uncertainty in estimates, and solve questions like How many runs for 1% accuracy?; 4. the above (three) also in some non-iid settings, e.g., for a stationary process, using the batch-means method; 5. for small-sized practical problems: build a suitable model and simulate it; 6. extract and collect required model parameters from a problem description or background; 7. recognize where modelling choices are made; 8. make elementary modelling decisions themselves; 9. recognize and describe in mathematical terms (the properties of) a number of stochastic processes, among others: Markov chains, random walks, Brownian motion; 10. recognize/identify the covered stochastic processes in practical settings/problems; 11. use the tools and concepts from probability theory to describe, discuss, and manipulate (the components of) stochastic models; 12. where applicable, solve problems by means other than Monte Carlo, i.e., analytically or numerically, procuring exact answers, by identifying a stochastic process and using its known properties; for example, concerning long-term system-behaviour.	
Education Method	Weekly class meeting of two hours. About two thirds spent on lecture; about one third on questions and exercises. A number of so-called "Hands-on" assignments: a collection of exercises that combine the analytical content with computer simulation; to be done individually or with a partner. The participants are expected to perform a significant amount of simulation in Python, Matlab, or R (or other suitable language). Spread over the semester, five homework assignments. I strive to make a classroom available during the (probably) Friday lunch break, in which the participants can work on their own and where the TA (if there is one) is available for questions. In addition, I will hold a weekly drop-in office hour, most likely at lunch time (details on Brightspace).	
Literature and Study Materials	Unfortunately, for this course there is no really appropriate textbook, one that combines modelling, Monte Carlo simulation, and stochastic processes. For the review of basic probability and statistics, you are referred to Dekking et al, "A modern introduction to probability and statistics," Springer, 2005; the text is very well suited for individual review and practice, as it contains many worked examples, (quick) exercises, answers and full solutions. To be successful in the course, participants must have developed superior skill wrt the tools and techniques in the first half of the book (see "Expected prior knowledge"). Most of the material on stochastic processes can be found in "Probability Models" by Sheldon M. Ross; the book contains many worked examples and exercises; its heuristic and intuitive approach fits that of the course; references will be to the 10th Edition. A slightly more mathematical, but still very accessible reference covering the same area is "An introduction to stochastic modeling" by Howard M. Taylor and Samuel Karlin. Lecture slides, notes, and additional references will be posted on Brightspace.	

	A laptop is required and it is recommended that you bring it to class. Many languages and/or software packages are suitable: I prefer you use the one you are most familiar with. I am migrating to Python, but in the material presented in class and distributed via Brightspace, there still is some Matlab code. You may use a language/tool other than those three, but talk to me first (because you should be able to use your software on the exam and it should be possible to arrange this). In all cases: you should have the software installed and be familiar with its use.
Assessment	1. Probability test on knowledge and skills needed in/for the course (15%); passing score must be obtained to take the course. 2. Assignments (five; each counts for 4%), which may be done with a (=one) partner. 3. Final exam (on-campus): closed-book computer exam (with cheatsheet allowed, specs on Brightspace); counts for 65%. To pass the course, the probability test must be passed and the final exam grade must be 6 or higher; course grade is weighted average.
Permitted Materials during Tests	To the final exam you may bring a self-compiled formula sheet: one A4 page with normal-sized text on about 25 lines (or any reasonable equivalent of this), made by you, of things you would find helpful.
Elective	Yes
Tags	Adventurous Diverse Mathematics Modelling Practicals Stochastics
Co-Instructor	Dr.ir. G.N.J.C. Bierkens

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	The final grade of the course consists of the following components: - Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). - Group programming project (group work) starting mid Q2 with submission deadline end Q2 (2/3 of final grade). Final grade calculation: (1/3 * homework assignments) + (2/3 * group programming project) but it is only valid when all homework programming assignments have been completed with sufficient results. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework programming assignments: submit new set of homework - Group programming project: submit improved version Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. J. Thies	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Robotics**

RO Second Year

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Robotics**

RO Electives 2nd year (Choose 1 out of 2, or 15 EC Free Electives)

IFM4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	To enter JIP, a student should be a 2nd-year master student; have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. be full time available (40 hours per week) in Q5; meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV Students will be selected by the JIP core staff based on motivation, study progress and recommendation.	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman,VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (Ilo;K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.	

	The minimum grade for the summative assessment is =/> 5.8. Process grade is 20% The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

RO57015	**RO Internship / Research Assignment**	**15**
Responsible Instructor	J.F.P. Kooij	
Instructor	Dr. A. Zgonnikov	
Contact Hours / Week x/x/x/x	x.x.0.0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The student will work as a starting Robotics Engineer in a professional research, design, or production environment, under the supervision of a professional expert. This can be a company, an academic institution, another external organisation, or TU Delft. The nominal working period is 15 EC (2.5 to 3 months full-time).	
Study Goals	The goal of the Internship / Research Assignment is to gain experience in working as an engineer at a company or an academic research environment. During the Internship / Research Assignment, the student learns how to: manage own scientific research independently plan and execute research in changing circumstances integrate new knowledge in an R&D project, considering ambiguity, incompleteness of information, and limitations work in a multidisciplinary team explain and defend outcomes from the research area to academia and industry act responsibly with regard to sustainability, economy, and social welfare.	
Education Method	Project	
Assessment	A report, to be assessed by your TUD supervisor. The Internship / Research Assignment is graded as 'Passed' (not a numeric grade) if the activities and the report meet the requirements. The report should provide sufficient view of your activities during the Internship / Research Assignment, the research or development performed, and your learning experience. There is no template for the report.	
Enrolment / Application	Students should approach a senior staff member in the Cognitive Robotics department as a potential supervisor to define a topic. The Internship / Research Assignment must be approved both by the supervisor and the Internship Coordinator. The supervisor and the coordinator will evaluate whether the topic is of a sufficient engineering level. The Internship / Research Assignment can be performed on the same topic as the MSc thesis; this is efficient and enhances depth. Currently, the Internship Coordinator is Arkady Zgonnikov <A.Zgonnikov@tudelft.nl>. Procedures are at: https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/study-abroad/internships-1. For Internships / Research Assignments in the Netherlands and abroad, start searching, respectively, at least two and three months before the intended due date. Always make sure you have the prior consent of one of the senior Cognitive Robotics staff members before making any commitments. The following criteria are applicable: The assignment should contain problem-solving activities (remember: the project is part of your engineering education). The assignment must be challenging and at MSc level. At the end, you should be able to report on your work, which means that you must have performed a comprehensive task. Contributing to a variety of not-related things in a company, however interesting, is not suitable for an Internship / Research Assignment. The field of work is the domain of the research of Robotics. ** Please check the FAQ Internship document on the MSc Robotics Brightspace page for further information. ** -- Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Cognitive Robotics	

Year	**2024/2025**
Organization	**Mechanical Engineering**
Education	**Master Robotics**

RO Graduation Project Obligatory (45 ECTS)

RO57010-20	RO Literature Research	10
Responsible Instructor	J.F.P. Kooij	
Instructor	K.J. van Tongeren	
Contact Hours / Week x/x/x/x	0.x.x.x	
Education Period	None (Self Study)	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The aim of the literature study is to learn how to independently search for recent scientific publications (i.e., articles, theses, books). The literature study is carried out individually. The topic of the literature study is linked to the graduation project. The findings from the literature study are used to motivate the research plan of the thesis and are instrumental for achieving a high-quality final thesis.	
Study Goals	After completing this course, students will be able to: study a topic by critically selecting relevant scientific literature identify and acquire lacking expertise identify the most interesting and relevant questions or problems in the field of the MSc thesis analyse and solve engineering problems in a systematic way present and report in good English	
Education Method	The literature study is individual work. The topic is chosen by the student together with the MSc thesis supervisor.	
Assessment	T1 Literature Research (written report; minimum grade: 6.0). The report is graded by the supervisor. T2 Presentation (colloquium) (pass/fail). The presentation is combined with the introductory phase of the MSc project. The colloquium is graded by an ad-hoc committee of scientific employees of the department. T3 Colloquium Attendance (pass/fail). All students must attend at least 10 colloquia.	
Department	ME Department Cognitive Robotics	

RO57010-20 toets 1	*Literature Research*	*10*

RO57010-20 toets 2	*Presentation*	*0*

RO57010-20 toets 3	*Colloquium Attendance*	*0*

RO57035	RO MSc Thesis	35
Responsible Instructor	J.F.P. Kooij	
Contact Hours / Week x/x/x/x	0.x.x.x	
Education Period	2 3 4	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The thesis work is the students individual masterpiece. Students have to dig deep into a research or development topic, show their theoretical knowledge and creativity, and generate new knowledge or develop a new design. They have regular meetings with their supervisor(s) and show their responsibility in work, writing, and time and project management. They have to write a well-structured scientific report, present their achievements in a public colloquium, and defend their work before a graduation committee. If a student undertakes the MSc thesis project at a company, a graduation agreement should be signed. In this respect, the template of the TUD-ME graduation agreement should be used, which is available at the following link: https://www.tudelft.nl/en/student/me-student-portal/education/related/student-forms/msc-forms	
Study Goals	After completing this course, students will be able to: analyse and solve engineering problems in a systematic way analyse problems and use modelling, simulation, design and integration towards solutions systematically design complex robotics systems generate knowledge within the discipline of Robotics generate innovative contributions to the discipline of Robotics apply paradigms, methods, and tools to design robotics systems design, conduct, and evaluate experiments design and control robotics systems in an interactive and noisy environment write a scientific report about ones own research manage own scientific research independently critically reflect on own knowledge, skills, and attitude take a standpoint with regard to a scientific argument within the research area work both independently and in multidisciplinary teams present and report in good English evaluate and assess the technological, ethical, and societal impact of own work	
Education Method	The MSc thesis project is the final part of the Robotics programme. The thesis work is carried out individually. Optionally, the project is undertaken in collaboration with the Robotics industry, in which case students will have a supervisor from the industry and graduation supervisor(s) from the TU Delft. The MSc thesis has to be completed as the last item of your course program. You cannot complete courses or your internship/research assignment/JIP after the MSc defence. Students prepare the MSc thesis as a research paper with appendices or as a project report.	
Assessment	Thesis, presentation, and defence. The criteria for grading the master thesis can be found here: https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/formulieren/msc-formulieren	
Enrolment / Application	You may start with your MSc Thesis when you have a minimum of 54 EC in first‐year courses (i.e., all your obligatory courses and most of your elective courses). Students should directly approach potential supervisors to define an MSc thesis topic. If you intend to do your MSc Thesis outside the TU Delft (e.g., at a company, another ME department, TUD faculty, or university), find a supervisor at CoR willing to supervise that thesis topic before making commitments to any external party. A graduation agreement has to be signed in all cases where a student graduates at a company. The graduation agreement can be found here: https://www.tudelft.nl/studenten/me-studentenportal/onderwijs/gerelateerd/formulieren/msc-formulieren. The COR supervisor will assess whether the assignment is of a sufficient academic level and whether it matches your prior knowledge. When you are almost ready with your MSc Thesis (and at least four weeks before the intended graduation date), your supervisor will organise a so-called green-light meeting. You will have to submit your concept thesis to your supervisor before this meeting, and your supervisor will assess (occasionally in consultation with the other members of the MSc Thesis Committee) whether your thesis is of sufficient level. If no green light is given, then a new green-light meeting is planned with an updated concept thesis. If green light is given, you should apply for the MSc exam. You should submit your report in paper and digital form (unless agreed otherwise) to the committee members no later than two weeks before the defence. ** Please see the Checklist Second Year on the MSc Robotics Brightspace page for further information. ** -- Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
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Department	Algorithmics

Dr.ir. P.A. Bouter

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Ir. M.B. Duinkerken

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Ir. J.B. Dönszelmann

Department	Computer Science & Engineering-Teaching Team

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